Banking under Conflict: Managers and Organizational Design

Appendix

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A Model details and full derivations

A.1 Environment, timing, and interpretation

A loan has verifiable size $L \in [\underline{L}, \bar{L}]$ and a mean-zero payoff-relevant state s (the optimal lending stance) with variance V. The party holding authority chooses a noncontractible lending stance y, and HQ incurs $L(y-s)^2$. Let g denote borrower ethnicity and $\pi_i \equiv \mathbb{P}(g=i)$ the share of borrowers coethnic with manager i; when the bank and city groups differ, $\pi_C > \pi_B$. Manager types have the same ability to acquire information from and bias toward in-group loan applicants; differences arise through π_i. Manager i has payoff

$$u_i(y, s, g, L) = -L\left[y - (s + b(\gamma)\mathbf{1}\{g=i\})\right]^2, \tag{a1}$$

where $b(\gamma) > 0$ is own-group bias; she is aligned with HQ on out-group loans. Loan size has continuous distribution F and mean $\mu \equiv \mathbb{E}[L]$.

Conflict satisfies

$$b'(\gamma) > 0, \qquad q'(\gamma) \leq 0, \qquad \chi^{O\prime}(\gamma) < \chi^{I\prime}(\gamma) \leq 0, \tag{a2}$$

so $\Delta'_\chi(\gamma) > 0$, where $\Delta_\chi(\gamma) \equiv \chi^I(\gamma) - \chi^O(\gamma) \geq 0$: cross-group information deteriorates more quickly than in-group information. The signal experiments below implement the information structure used in the main text.

The framework combines standard models of delegation (Philippe Aghion and Jean Tirole, 1997; Ricardo Alonso, Wouter Dessein and Niko Matouschek, 2008; Wouter Dessein, 2002) with work on hierarchy and soft information in banking (Jose M. Liberti and Atif R. Mian, 2009; Jeremy C. Stein, 2002). The action y is the residual underwriting stance—for example, how soft evidence is translated into pricing, collateral, monitoring, or other terms—after verifiable rules have been imposed, and L affects only the stakes of that decision. The benchmark assumes that borrower ethnicity g is independent of L and s, and that residual uncertainty and conditional signal quality do not vary with L, producing a single size-based approval cutoff. HQ maximizes bank value. Managers of any ethnicity may favor their own group. The main-text comparison uses the bank-coethnic type B and city-coethnic type C, but the same expressions apply to any candidate i, including a third-group manager N. For conflict comparative statics, (π_i, F, κ) are fixed; for composition comparative statics, the conditional information and preference primitives are fixed.

Information acquisition is match-dependent. Shared language, networks, and cultural codes make it easier to elicit and interpret a coethnic borrower's information (Raymond Fisman, Daniel Paravisini and Vikrant Vig, 2017), while an out-group interaction can still generate recordable evidence. Only an own-group match generates the additional soft information assessment used in the communication problem, and the same match can generate favoritism (Raymond Fisman, Arkodipta Sarkar, Janis Skrastins and Vikrant Vig, 2020). Once elicited, recordable information enters the loan file and survives either authority regime; soft information is privately observed by the manager. Conflict γ is a reduced-form salience index, empirically proxied by network exposure rather than necessarily local casualties. It can strengthen identity and disrupt trust, disclosure, and the interpretation of evidence across ethnic boundaries. The key asymmetry is therefore generated by a common match-level technology: cross-group recordable information deteriorates more rapidly than in-group recordable information, and the bank-coethnic manager faces the cross-group technology on a larger borrower share.

HQ appoints a manager and commits before loan information is realized to a coarse authority schedule based on manager type and verifiable loan size, but not borrower ethnicity or realized information. The

restriction reflects branch-wide approval limits rather than borrower-specific intervention. If authority could depend on ethnicity, HQ could review only loans from applicants coethnic with the manager and π_i would drop out of the cutoff. Under review, HQ observes the recordable assessment on every loan, receives a report about the soft-information assessment on an own-group loan, chooses y, and incurs the fixed cost κ; because the benefit of review scales with L while its cost does not, authority takes a loan-size cutoff form. Formal approval rights are contractible, but y and the report are not, so HQ cannot commit to a report-contingent action. We use pure-strategy perfect Bayesian equilibria in the monotone-partition class of Vincent P. Crawford and Joel Sobel (1982); Section A.4 proves that equilibrium selection does not affect the authority rule.

A.2 A feasible match-dependent information structure

The benchmark takes the conditional residual-state distribution and signal experiments, given the manager–borrower match, to be invariant across loan sizes. For feasibility, let $V_x, V_t \geq 0$ and $V_\varepsilon > 0$, with $V = V_x + V_t + V_\varepsilon$, and suppose

$$0 \leq \chi^O(\gamma) \leq \chi^I(\gamma) \leq V_x, \qquad 0 \leq q(\gamma) \leq \sqrt{3V_t} \tag{a3}$$

for every relevant conflict level.
Let the state decompose as

$$s = r + \tau + \varepsilon, \tag{a4}$$

where r, τ, and ε are mutually independent, have mean zero, and satisfy

$$\mathrm{Var}(r) = V_x, \qquad \mathrm{Var}(\tau) = V_t, \qquad \mathrm{Var}(\varepsilon) = V_\varepsilon > 0. \tag{a5}$$

For concreteness, let $r \sim U[-\sqrt{3V_x}, \sqrt{3V_x}]$ and $\tau \sim U[-\sqrt{3V_t}, \sqrt{3V_t}]$, with the natural degenerate interpretation when a component has zero variance. The component r represents information that can be elicited and recorded in a loan file; τ represents soft information whose posterior assessment must be communicated.
For match status $m \in \{I, O\}$, where I denotes an own-group borrower and O an out-group borrower, the recordable experiment produces

$$x_i^m = \mathbb{E}[r \mid \mathcal{R}_i^m], \qquad x_i^m \sim U[-p^m, p^m], \qquad p^m \equiv \sqrt{3\chi^m}. \tag{a6}$$

The assessment x_i^m is verifiable once elicited and observed by HQ. On an own-group loan, a second experiment produces

$$t_i = \mathbb{E}[\tau \mid \mathcal{T}_i], \qquad t_i \sim U[-q, q], \tag{a7}$$

which is privately observed by the manager. An out-group loan generates no soft information.

Lemma 1 (Bayesian feasibility)**.** *For every (χ^I, χ^O, q) satisfying (a3), signal experiments satisfying (a6)–(a7) exist. Their posterior means satisfy*

$$\mathbb{E}[s \mid x_i^I, t_i, g = i] = x_i^I + t_i, \qquad \mathbb{E}[s \mid x_i^O, g \neq i] = x_i^O. \tag{a8}$$

The state remains non-invertible because $V_\varepsilon > 0$.

Proof. Let $Z \sim U[-1,1]$. For $0 \le a \le A$, write $a = \lambda A$ with $\lambda \in [0,1]$. For every convex function φ,

$$\varphi(aZ) = \varphi(\lambda AZ + (1-\lambda)0) \le \lambda\varphi(AZ) + (1-\lambda)\varphi(0).$$

Taking expectations and applying Jensen's inequality gives $\mathbb{E}[\varphi(aZ)] \le \mathbb{E}[\varphi(AZ)]$. Thus $U[-a,a]$ is dominated by $U[-A,A]$ in convex order. By the martingale-coupling theorem of Volker Strassen (1965), there is a signal experiment whose posterior mean has distribution $U[-a,a]$. Apply this result to r with $(a,A) = (p^m, \sqrt{3V_x})$ for each $m \in \{I,O\}$ and to τ with $(a,A) = (q, \sqrt{3V_t})$. Independence of the components and experiments yields (a8); ε retains positive variance. □

Calibration and orthogonality imply residual loss

$$\mathbb{E}[(s - x_i^I - t_i)^2 \mid g = i] = V - \chi^I - \frac{q^2}{3}, \tag{a9}$$

$$\mathbb{E}[(s - x_i^O)^2 \mid g \ne i] = V - \chi^O. \tag{a10}$$

Define portfolio-average recordable information

$$\bar{\chi}_i = (1-\pi_i)\chi^O + \pi_i\chi^I = \chi^O + \pi_i\Delta_\chi. \tag{a11}$$

Averaging the residual losses gives

$$I_i = V - \bar{\chi}_i - \pi_i\frac{q^2}{3}, \tag{a12}$$

which is the residual information loss entering both authority regimes. The city-coethnic manager's recordable-information advantage follows without type-specific signal technologies:

$$\bar{\chi}_C - \bar{\chi}_B = (\pi_C - \pi_B)\Delta_\chi \ge 0. \tag{a13}$$

Total information is likewise better under C whenever $\pi_C > \pi_B$ and $\Delta_\chi + q^2/3 > 0$.

A.3 Delegated decisions

On an own-group loan, manager i observes (x_i^I, t_i) and minimizes

$$\mathbb{E}\left[(y - s - b)^2 \mid x_i^I, t_i, g = i\right].$$

By (a8), her delegated action is

$$y_i^{\text{del}} = x_i^I + t_i + b. \tag{a14}$$

HQ's expected loss is $V - \chi^I - q^2/3 + b^2$. On an out-group loan, the manager is aligned and chooses $y = x_i^O$, generating loss $V - \chi^O$. Averaging across borrower groups yields

$$\begin{aligned}\ell_i^{\text{del}}(L) &= \left[(1-\pi_i)(V-\chi^O) + \pi_i\left(V - \chi^I - \frac{q^2}{3} + b^2\right)\right]L \\ &= \left(I_i + \pi_i b^2\right)L,\end{aligned} \tag{a15}$$

This is the expected loss under delegation.

A.4 Biased communication of soft information

Consider an own-group borrower under manager i. The recordable assessment x_i^I is observed by HQ. Conditional on x_i^I, HQ's ideal residual action is t_i, while the manager's ideal residual action is $t_i + b$. The communication problem therefore depends only on the privately observed soft-information component $t_i \sim U[-q, q]$. If $q = 0$, this component is degenerate, no informative communication is possible, and the review loss below follows directly. For the partition derivation, take $q > 0$.

We characterize the standard pure-strategy monotone partition equilibria. Let

$$-q = z_0 < z_1 < \cdots < z_K = q$$

be the boundaries of a K-bin equilibrium and let $d_k = z_k - z_{k-1}$. Since t_i is uniform, HQ chooses the midpoint

$$m_k = \frac{z_{k-1} + z_k}{2}$$

following a report corresponding to bin k. At an interior boundary z_k, sender indifference requires

$$[m_k - (z_k + b)]^2 = [m_{k+1} - (z_k + b)]^2,$$

which is equivalent to

$$d_{k+1} = d_k + 4b. \tag{a16}$$

Using $\sum_{k=1}^{K} d_k = 2q$ gives

$$d_k = \frac{2q}{K} + 2b(2k - K - 1). \tag{a17}$$

The smallest bin is d_1, so a genuine K-bin equilibrium exists if and only if

$$bK(K-1) < q. \tag{a18}$$

The one-bin uninformative equilibrium always exists.

Conditional on a bin of length d_k, HQ's mean-squared error in predicting t_i is $d_k^2/12$. The ex ante communication loss is

$$\begin{aligned} C_K(b, q) &= \sum_{k=1}^{K} \frac{d_k}{2q} \frac{d_k^2}{12} \\ &= \frac{q^2}{3K^2} + \frac{b^2}{3}(K^2 - 1). \end{aligned} \tag{a19}$$

For every informative monotone partition equilibrium, $K \geq 2$. By (a18),

$$\begin{aligned} \frac{C_K}{b^2} &> \frac{(K-1)^2 + K^2 - 1}{3} \\ &= \frac{2K(K-1)}{3} > 1. \end{aligned} \tag{a20}$$

Thus every informative equilibrium in this class creates a larger decision distortion than delegation, even before the review cost is paid, and is never chosen by HQ.

For $K = 1$, HQ learns nothing about t_i, so

$$C_1 = \mathrm{Var}(t_i) = \frac{q^2}{3}.$$

If uninformative review is preferred to delegation for any loan, then necessarily

$$b^2 > \frac{q^2}{3},$$

which implies $b > q/\sqrt{3} > q/2$. A two-bin equilibrium requires $2b < q$, and equilibria with more bins require still smaller bias. Hence, whenever review can be optimal, no informative monotone partition equilibrium exists.

Under uninformative review, HQ retains x_i^I but not t_i. Expected loss on an own-group loan is therefore

$$V - \chi^I, \tag{a21}$$

which exceeds the full-information residual by $q^2/3$. On an out-group loan, x_i^O is recordable and no strategic report is required, so loss is $V - \chi^O$. Averaging over borrower groups and adding κ gives

$$\begin{aligned} \ell_i^{\mathrm{hq}}(L) &= \left[(1-\pi_i)(V-\chi^O) + \pi_i(V-\chi^I)\right] L + \kappa \\ &= \left(I_i + \pi_i \frac{q^2}{3}\right) L + \kappa, \end{aligned} \tag{a22}$$

This is the relevant review loss whenever review can be optimal.

Lemma 2 (Selection robustness)**.** *Within pure-strategy monotone-partition perfect Bayesian equilibria, the optimal authority rule is invariant to equilibrium selection. If an informative partition equilibrium exists, delegation strictly dominates review for HQ; if review is optimal, no informative partition equilibrium exists and the relevant review outcome is uninformative.*

Proof. For any $K \geq 2$, (a20) gives $C_K > b^2$ before the review cost is paid. If uninformative review is weakly preferred for some loan, then $b^2 > q^2/3$, which implies $2b > q$ and violates (a18) for every $K \geq 2$. □

A.5 The authority rule and comparative statics

For the derivations, define

$$d(\gamma) \equiv b(\gamma)^2 - \frac{q(\gamma)^2}{3}, \qquad h_i(\gamma) \equiv \pi_i d(\gamma). \tag{a23}$$

and write the delegation indicator, delegated share, and delegated loan-size moment as

$$D_i(L,\gamma) = \mathbf{1}\{\pi_i d(\gamma) L \leq \kappa\}, \qquad A_i = \mathbb{E}[D_i], \qquad M_{D,i} = \mathbb{E}[L D_i]. \tag{a24}$$

Subtracting (a22) from (a15) shows that delegation is weakly preferred precisely when

$$\pi_i \left(b^2 - \frac{q^2}{3}\right) L \leq \kappa. \tag{a25}$$

This proves the authority rule and (2). To nest the full-delegation corner, define

$$c_i(\gamma) = \begin{cases} +\infty, & h_i(\gamma) \le 0, \\ \widehat{L}_i^*(\gamma), & h_i(\gamma) > 0. \end{cases} \tag{a26}$$

Then $D_i(L,\gamma) = \mathbf{1}\{L \le c_i(\gamma)\}$. Portfolio-average recordable information $\bar{\chi}_i$ cancels because it remains available under both authority regimes.

The authority conclusions require only that the net gain from review on an own-group match increases, $d'(\gamma) > 0$. The primitive conditions in (a2) are sufficient because

$$d'(\gamma) = 2bb' - \frac{2}{3}qq' > 0. \tag{a27}$$

For any manager with $\pi_i > 0$,

$$h_i'(\gamma) = \pi_i d'(\gamma) > 0. \tag{a28}$$

Therefore $D_i(L,\gamma)$ is weakly decreasing in conflict for every L, including changes that enter or leave a corner. Whenever $h_i > 0$,

$$\frac{\mathrm{d}\widehat{L}_i^*}{\mathrm{d}\gamma} = -\frac{\widehat{L}_i^*}{h_i} h_i' = -\widehat{L}_i^* \frac{d'}{d} < 0. \tag{a29}$$

Extend F by zero below $\underline{L}$, one above $\bar{L}$, and set $F(+\infty) = 1$. Then $A_i(\gamma) = F(c_i(\gamma))$. For $\gamma_2 > \gamma_1$,

$$A_i(\gamma_2) - A_i(\gamma_1) = F(c_i(\gamma_2)) - F(c_i(\gamma_1)) \le 0. \tag{a30}$$

The inequality is strict if and only if

$$\mathbb{P}(c_i(\gamma_2) < L \le c_i(\gamma_1)) > 0. \tag{a31}$$

Continuity of F is not needed for this monotonicity or strictness statement.

For the composition comparative static, hold b and q fixed and hence hold d fixed. If $d \le 0$, then $h_i = \pi_i d \le 0$ for every π_i, so every loan is delegated and authority is constant in π_i. If $d > 0$, h_i increases linearly in π_i and, for $\pi_i > 0$,

$$\frac{\partial \widehat{L}_i^*}{\partial \pi_i} = -\frac{\widehat{L}_i^*}{\pi_i} < 0, \qquad \frac{\partial \widehat{L}_i^*}{\partial \kappa} = \frac{\widehat{L}_i^*}{\kappa} > 0. \tag{a32}$$

Thus, whenever $d > 0$, the cutoff $c_i = \widehat{L}_i^*$ strictly decreases in π_i for every $\pi_i > 0$. The pointwise delegation rule is weakly decreasing, and aggregate autonomy decreases strictly only if a positive mass of loans crosses the cutoff.

A.6 Appointment and the cross-group information restriction

For the bank–city comparison, suppose the groups differ, so $\pi_C > \pi_B$, and define $\Phi(\gamma) \equiv \Lambda_B(\gamma) - \Lambda_C(\gamma)$. For each loan size, HQ selects the smaller of the additional delegated loss $\pi_i b^2 L$ and the reviewed loss $\pi_i q^2 L/3 + \kappa$, on top of the residual information loss $I_i L$. Hence

$$\Lambda_i(\gamma) = I_i(\gamma)\mu + \mathbb{E}\left[\min\left\{\pi_i b(\gamma)^2 L, \pi_i \frac{q(\gamma)^2}{3} L + \kappa\right\}\right]. \tag{a33}$$

Factoring out the soft-information term gives (3) in the main text. The exact appointment decomposition is

$$\begin{aligned}\Phi(\gamma) &= (\bar{\chi}_C - \bar{\chi}_B)\mu + \mathbb{E}\left[\min\{h_B L, \kappa\} - \min\{h_C L, \kappa\}\right] \\ &= (\pi_C - \pi_B)\Delta_\chi \mu + \mathbb{E}\left[\min\{h_B L, \kappa\} - \min\{h_C L, \kappa\}\right]. \end{aligned} \tag{a34}$$

The first term is C's recordable-information advantage: she acquires more precise information from the, relatively larger, share of borrowers coethnic with her. The second term captures the loss due to the incentive issue this creates.

Because F is continuous, the equality event $h_i L = \kappa$ has probability zero. Wherever the primitives are differentiable, the envelope theorem gives

$$\Lambda_i' = -\bar{\chi}_i' \mu + h_i' M_{D,i}. \tag{a35}$$

Since $\bar{\chi}_i' = \chi^{O\prime} + \pi_i \Delta_\chi'$, the common deterioration in recordable information cancels from the appointment comparison, leaving

$$\Phi' = \mu(\pi_C - \pi_B)\Delta_\chi' + h_B' M_{D,B} - h_C' M_{D,C}. \tag{a36}$$

Thus conflict shifts appointment toward C if and only if

$$\mu(\pi_C - \pi_B)\Delta_\chi' > h_C' M_{D,C} - h_B' M_{D,B}. \tag{a37}$$

The left side is C's relative information advantage as cross-group borrower communication deteriorates: because $\pi_C > \pi_B$, she relies on the less resilient out-group technology for a smaller portfolio share. The right side is the relative increase in the minimized loss associated with delegation and review. Here cross-group communication means borrower–manager information acquisition, not manager–HQ cheap talk. Unlike a direct assumption on χ_B' and χ_C', the asymmetry is generated by one common match-level technology and the observed difference $\pi_C > \pi_B$.

A primitive dominance condition: Under the maintained common own-group bias and soft-information technologies, $d = b^2 - q^2/3$, $h_i' = \pi_i d'$, and

$$\Phi' = \mu(\pi_C - \pi_B)\Delta_\chi' - d'\left[\pi_C M_{D,C} - \pi_B M_{D,B}\right]. \tag{a38}$$

For every loan size,

$$\pi_C L D_C - \pi_B L D_B \leq (\pi_C - \pi_B) L. \tag{a39}$$

If $d \leq 0$, both types delegate and the inequality holds with equality. If $d > 0$, $D_C \leq D_B$: the only additional case has $D_B = 1$ and $D_C = 0$, making the left side negative. Taking expectations in (a39) yields

$$\Phi' \geq \mu(\pi_C - \pi_B)(\Delta_\chi' - d'). \tag{a40}$$

Consequently,

$$\Delta_\chi' > d' \tag{a41}$$

is sufficient for conflict to favor C under any optimal authority allocation. If all loans are delegated, (a40) holds with equality, so (a41) is also necessary. The comparison poses the model's empirical appointment question: does conflict widen the in-group recordable-information premium faster than it raises the net gain

from reviewing an own-group loan? When $q' = 0$, it reduces to $\Delta'_\chi > 2bb'$; allowing $q' \leq 0$ replaces $2bb'$ with $d' = 2bb' - 2qq'/3$ and leaves the authority result unchanged.

Corollary (multi-type appointment ordering): Let j and k be any candidate manager ethnicities with $\pi_j > \pi_k$. Repeating the pairwise argument above gives

$$\begin{aligned}\Lambda'_j - \Lambda'_k &= -\mu(\pi_j - \pi_k)\Delta'_\chi + d' \left[\pi_j M_{D,j} - \pi_k M_{D,k}\right] \\ &\leq -\mu(\pi_j - \pi_k)(\Delta'_\chi - d').\end{aligned} \tag{a42}$$

Hence (a41) makes every higher-π candidate relatively more attractive as conflict rises, absent differential assignment-cost responses. In particular, any third-group manager N with $\pi_N > \pi_B$ becomes more attractive relative to B; if $\pi_C > \pi_N > \pi_B$, the operating-loss responses satisfy $\Lambda'_C < \Lambda'_N < \Lambda'_B$. The result does not sign the aggregate share of N, because N can gain appointments from B while losing them to candidates with still larger π_i.

A.7 Managerial assignment costs

Let $w_i(\gamma)$ be a differentiable reduced-form cost of assigning manager i, measured in the same branch-level units as Λ_i and encompassing relocation, retention, or safety premia. Define $\Omega_i \equiv \Lambda_i + w_i$. HQ chooses the candidate with the lowest Ω_i. Define $\Psi(\gamma) \equiv \Omega_B(\gamma) - \Omega_C(\gamma) = \Phi(\gamma) + w_B(\gamma) - w_C(\gamma)$. HQ appoints C if and only if $\Psi \geq 0$, and conflict shifts appointment toward C if and only if

$$\mu(\pi_C - \pi_B)\Delta'_\chi + (w'_B - w'_C) > h'_C M_{D,C} - h'_B M_{D,B}. \tag{a43}$$

Thus a rise in the relative relocation, retention, or safety cost of assigning B makes reallocation toward C more likely. A conflict-induced customer preference for a city-coethnic manager enters identically as a reduction in w_C. Appointment alone identifies only the composite condition (a43); it does not disentangle information dominance from assignment-cost responses. Auxiliary tests based on reassigned insiders, authority changes holding manager type fixed, and managers' local origins would bear more directly on the cross-group information interpretation.

Conditional on the appointed manager, w_i cancels from the authority comparison $h_i L \leq \kappa$, leaving Parts 2 and 3 of Proposition (1) unaffected. Assignment costs can nevertheless change observed branch-level authority by changing the appointed type. Holding manager type fixed removes this selection channel and provides the cleanest empirical analogue of the direct information–incentive response.

A.8 Formal comparative statics and proof

Proposition (1) states these results in words. The following is its formal version, using the notation in (a23) and (a24). "Conflict favors C" means $\Psi' > 0$; it need not cause a switch unless the appointment threshold is crossed.

Proposition 1 (Appointment and authority: formal statement)**.** *Suppose the primitives are differentiable,* (π_i, F, κ) *are fixed in conflict,* (a2) *holds, and* F *is continuous; for Part 1, suppose the bank and city groups differ, so* $\pi_C > \pi_B$*.*

(i). Conflict favors C if and only if

$$\mu(\pi_C - \pi_B)\Delta'_\chi + (w'_B - w'_C) > d'\left(\pi_C M_{D,C} - \pi_B M_{D,B}\right). \tag{a44}$$

With equal assignment-cost responses, the condition

$$\Delta'_\chi(\gamma) > d'(\gamma) \tag{a45}$$

is sufficient in general and necessary and sufficient when both candidate types would delegate all loans.

(ii). Conflict weakly reduces authority conditional on manager type. For $\pi_i > 0$ and $\gamma_2 > \gamma_1$, $D_i(L, \gamma_2) \leq D_i(L, \gamma_1)$ for every L; whenever $h_i > 0$, $\widehat{L}_i^*$ strictly decreases.

(iii). Holding b and q fixed, a larger coethnic customer share weakly lowers authority. If $d > 0$, $\widehat{L}_i^* = \kappa/(\pi_i d)$ strictly decreases in $\pi_i > 0$; if $d \leq 0$, all loans remain delegated.

Derivation. The envelope theorem gives

$$\Psi' = \mu(\pi_C - \pi_B)\Delta'_\chi + (w'_B - w'_C) - d'(\pi_C M_{D,C} - \pi_B M_{D,B}).$$

Because $\pi_C > \pi_B$ implies $D_C \leq D_B$, $\pi_C M_{D,C} - \pi_B M_{D,B} \leq (\pi_C - \pi_B)\mu$, proving Part 1. Parts 2 and 3 follow from $d' = 2bb' - 2qq'/3 > 0$ and $h_i = \pi_i d$; aggregate authority changes strictly only when positive loan mass crosses the cutoff. □

Corollary (appointment ordering). For any candidates j and k with $\pi_j > \pi_k$, condition (a45) and equal assignment-cost responses imply

$$\frac{\mathrm{d}}{\mathrm{d}\gamma}\left(\Omega_j - \Omega_k\right) < 0. \tag{a46}$$

Thus conflict favors candidates with larger local borrower shares. In particular, any third-group manager N with $\pi_N > \pi_B$ becomes more attractive relative to B; whether N is appointed depends on assignment costs and on π_N relative to the other candidates.

A.9 Appointment and restricted authority can coexist

The recordable information premium allows HQ to appoint an informed local manager while reviewing some of her decisions. Review preserves the recordable assessment even when strategic communication eliminates the value of the soft component.

Lemma 3 (Nonempty coexistence region)**.** *A sufficient condition for C to be preferred to B, including assignment costs, is*

$$\left[\bar{\chi}_C - \bar{\chi}_B - \pi_B \frac{q^2}{3}\right]\mu > \kappa + w_C - w_B. \tag{a47}$$

Under the common match-level technology, the bracket equals $(\pi_C - \pi_B)\Delta_\chi - \pi_B q^2/3$. A sufficient condition for HQ to review a positive mass of decisions under C is

$$h_C > 0 \quad \textit{and} \quad \mathbb{P}\left(L > \frac{\kappa}{h_C}\right) > 0. \tag{a48}$$

These conditions hold jointly for a nonempty parameter region.

Proof. The minimized agency and review term in Λ_B is nonnegative, so

$$\Lambda_B \geq I_B \mu = \left(V - \bar{\chi}_B - \pi_B \frac{q^2}{3} \right) \mu.$$

Under C, reviewing every loan is feasible. Hence

$$\Lambda_C \leq (V - \bar{\chi}_C)\, \mu + \kappa.$$

Adding assignment costs to these bounds gives

$$\Omega_B - \Omega_C \geq \left[\bar{\chi}_C - \bar{\chi}_B - \pi_B \frac{q^2}{3} \right] \mu - \kappa + w_B - w_C.$$

Condition (a47) therefore implies $\Omega_C < \Omega_B$. Condition (a48) places positive loan mass above C's approval cutoff. To see that the two conditions can hold jointly, take equal assignment costs, a positive feasible Δ_χ, sufficiently small q and κ, and sufficiently large b. HQ can then appoint C for her recordable information while reviewing a positive mass of her decisions. □

A positive recordable match premium separates the informational value of appointment from the informational cost of review. If $\Delta_\chi = 0$ and all of C's relative information advantage is in soft information, reviewing own-group loans destroys that advantage and the tension between appointment and centralization can reappear.

B Inference and bank-time comparisons

B.1 Manager ethnicity and delegation with bank-time fixed effects

Table B1 compares branches in bank-coethnic and non-bank-coethnic cities within the same bank and period. For this exercise, Network Exposure is deliberately measured over the bank's complete branch network, including conflict near the focal branch. It therefore varies only at the bank-time level and its uninteracted main effect is absorbed by bank-time fixed effects. Panel A reports coefficients on the interaction between this exposure and an indicator for a non-bank-coethnic city. The four outcomes are the three manager types and delegation. All specifications include branch and bank-time fixed effects and allow non-bank-coethnic branches to follow a separate time trend. The city-coethnic manager estimate is 0.128 (s.e. 0.030), while the delegation estimate is $-2{,}543$ thousand 2018 ETB (s.e. 934). Panel B relates delegation to manager ethnicity and the local population share belonging to the manager's ethnic group separately in bank-city non-coethnic and coethnic locations.

Table B1: Manager ethnicity and delegation with bank-time fixed effects

	Panel A: Manager ethnicity and delegation			
	(1) City-coethnic manager	(2) Bank-coethnic manager	(3) Neither-coethnic manager	(4) Delegation
Network Exposure × Non-coethnic city	0.128***	-0.074	-0.012	-2543**
	(0.030)	(0.060)	(0.036)	(934)
Branch FE	Yes	Yes	Yes	Yes
Bank-time FE	Yes	Yes	Yes	Yes
Non-bank-coethnic-time FE	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.411	0.729	0.171	1755
N (Branches)	978	978	978	553
Banks	16	16	16	16

	Panel B: Delegation, manager type and local population share					
	Bank and city not coethnic			Bank and city coethnic		
	(1) Manager type	(2) + population share	(3) Population share	(4) Manager type	(5) + population share	(6) Population share
City-coethnic manager	-473.3	1211.0		-2423.9**	-767.6	
	(1012.2)	(1544.9)		(1041.8)	(1494.4)	
Neither-coethnic manager	736.6	944.2				
	(1404.2)	(1502.2)				
Manager-group population share		-4147.1**	-3200.4***		-3182.1	-4067.5**
		(1894.3)	(1051.9)		(1901.5)	(1480.1)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank-time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	1688.5	1688.5	1688.5	1849.3	1849.3	1849.3
N (Branches)	321	321	321	229	229	229
Banks	15	15	15	13	13	13

Notes: Panel A uses the whole-network measure of conflict exposure. This measure includes conflict near the focal branch and is constant within bank-time cells, so its main effect is absorbed. The reported coefficients are on the interaction between Network Exposure and an indicator for a branch in a city whose ethnicity differs from the bank's. The first three outcomes are indicators; delegation in column 4 is measured in thousands of 2018 ETB. Panel A includes non-bank-coethnic-time fixed effects to allow the two groups to follow different time trends. Panel B regresses delegation on manager type and the local population share belonging to the manager's ethnic group. Columns 1–3 report results for the non-bank-coethnic sample, and columns 4–6 report results for the bank-city-coethnic sample. Within each group, the columns include manager-type indicators, manager type plus the population share, and the population share alone. City-coethnic manager and Neither-coethnic manager are indicators. In columns 1–2, bank-coethnic managers are the omitted category. In columns 4–5, the bank's and city's ethnicities coincide, so managers coethnic with neither are the omitted category. Delegation is the largest loan a manager can approve without headquarters' approval, measured in thousands of 2018 ETB and winsorized at the 95th percentile within cells defined by time and bank-city coethnicity. All specifications include branch and bank-time fixed effects. Standard errors are clustered by bank and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

B.2 Delegation results for branches with no change in manager type

Table B2: Delegation among branches with an unchanged manager type

	(1) All stable types	(2) Bank or city coethnic	(3) Bank or city coethnic	(4) Bank or city coethnic
Panel A: Conflict exposure				
Network Exposure	-811.1	-749.7	-328.8	-442.1
	(539.9)	(567.8)	(1088.2)	(1075.6)
Network Exposure × City-coethnic			-691.6	
			(1100.3)	
Network Exposure × Population share				-614.6
				(1321.4)
2018 mean dep. var.	1709.7	1700.0	1700.0	1700.0
Branch-years	248	240	240	240
Branches	124	120	120	120
Panel B: 2018–2022 difference				
2022	6565.1***	6781.4***	8868.0***	10307.2***
	(838.6)	(859.3)	(1236.8)	(1417.5)
2022 × City-coethnic			-5007.7***	
			(1582.9)	
2022 × Population share				-8628.1***
				(1932.0)
2018 mean dep. var.	1709.7	1700.0	1700.0	1700.0
Branch-years	248	240	240	240
Branches	124	120	120	120

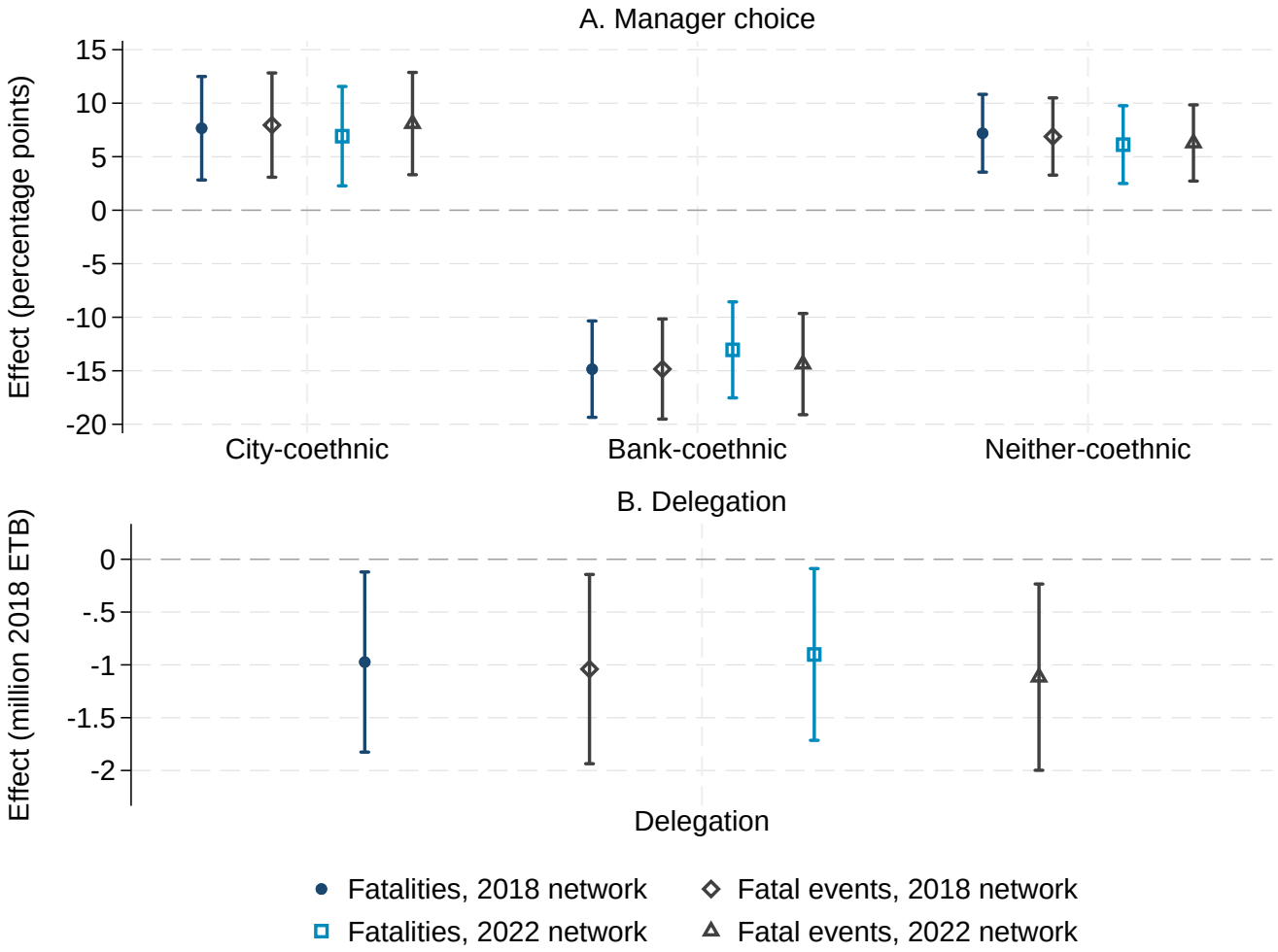
Notes: The sample is branch-years in cities not coethnic with the bank, restricted to branches whose manager type is unchanged between 2018 and 2022. The dependent variable in both panels is the largest loan a branch manager can approve without headquarters' approval, measured in thousands of 2018 ETB and winsorized as in main-text Table 3. *Panel A* reports the effect of network conflict exposure, the log intensity of branch-weighted exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest. *Panel B* reports the difference between the two waves and uses no conflict measure; it asks whether the two groups of branches diverged at all, rather than whether they diverged in proportion to realized exposure. Column 1 includes all stable manager types and reproduces the estimate quoted in the main text. Columns 2–4 restrict to branches whose manager is stably coethnic with either the bank or the city, excluding the four branches whose manager is coethnic with neither. Column 3 interacts the regressor with an indicator for a city-coethnic manager. Column 4 interacts it with the 2007 census population share of the manager's own ethnic group, the empirical counterpart of π_i; these are population shares rather than customer shares, since the survey records no customer ethnicity. Because manager type and the population share are time-invariant within branch in this sample, their levels are collinear with the branch fixed effects and are not estimable here; only the interactions reported are identified. All columns include branch fixed effects, and Panel A additionally includes wave fixed effects. Standard errors clustered at the branch level are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C Robustness Tests

C.1 Alternative conflict intensity and branch-network definitions

Figure C1 implements the two alternative definitions described in the main text for both manager choice and delegation. We replace reported fatalities with the number of distinct fatal ACLED events and, separately, replace the fixed 2018 branch network with the fixed network of branches observed in 2022. The fourth series changes both choices at once. To compare estimates across differently scaled exposures, the figure reports effects of a one-standard-deviation within-branch change in each log exposure. The appointment results are stable across all four definitions: network conflict raises the probability of a city-coethnic or neither-coethnic manager and lowers the probability of a bank-coethnic manager. Delegation also falls under all four definitions.

Figure C1: Manager choice and delegation under alternative conflict and network definitions

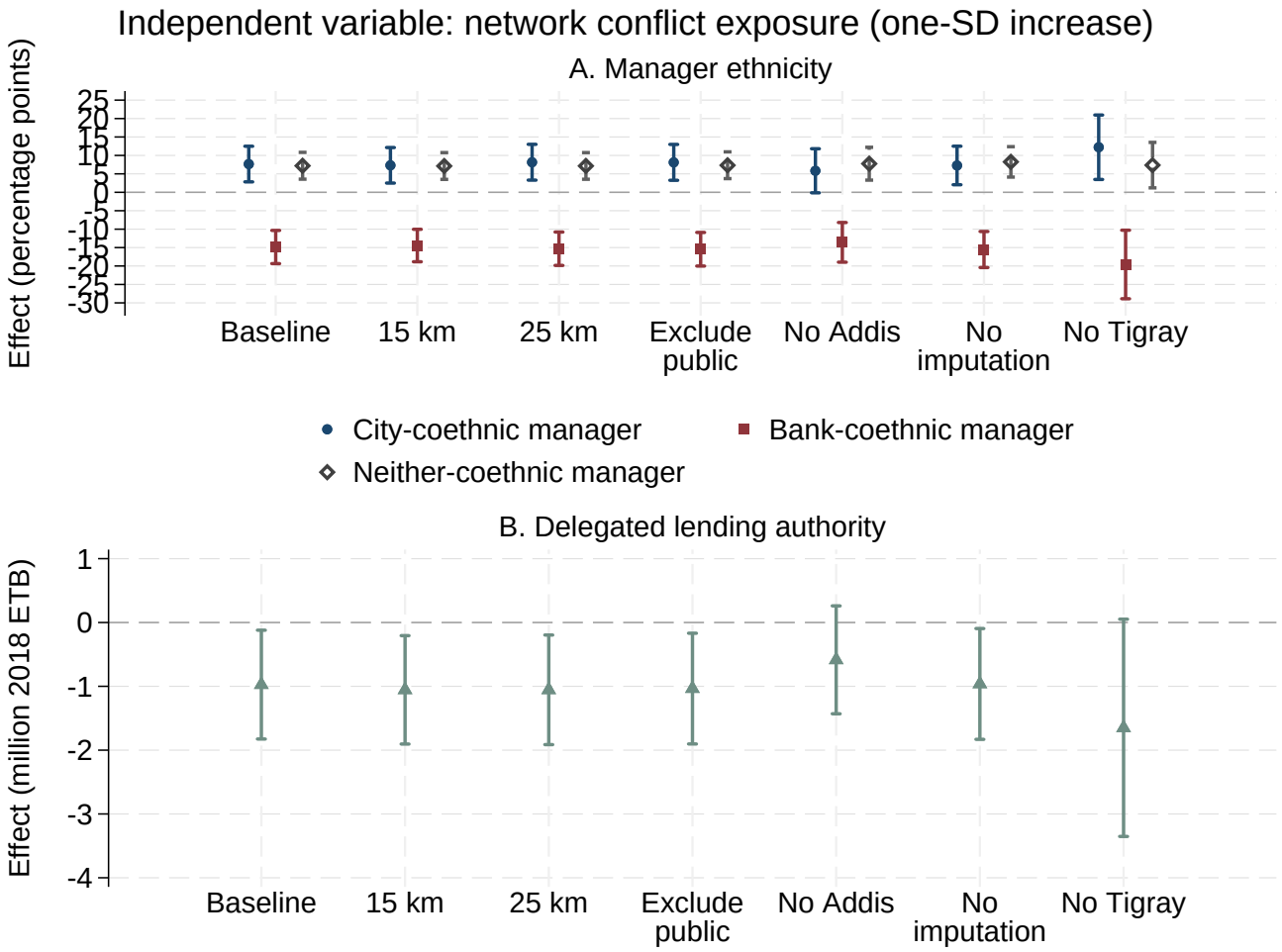


Notes: Panel A reports separate regressions for the three mutually exclusive manager-ethnicity outcomes among branches in non-bank-coethnic cities. Panel B reports delegation in millions of 2018 ETB. The four series form a 2×2 comparison of conflict intensity (reported fatalities or the number of distinct ACLED events with at least one reported fatality) and network vintage (the fixed 2018 or fixed 2022 survey branch network). The network for each vintage is held fixed across both conflict periods; the 2022 variant therefore changes the network footprint and branch weights, not the network contemporaneously with conflict. All definitions use the same actor–ethnicity mapping, 20 km event-to-city match, branch-count weights, 2015–2018 and 2019–2022 windows, denominator equal to the number of bank branches outside the focal city, and strict exclusion of every event within 20 km of the focal city. Exposure is transformed as $\ln(x + 0.01)$; no inverse-hyperbolic-sine transformation is used. Each regression includes branch and wave fixed effects. Standard errors are clustered by branch. Markers and 95% confidence intervals are scaled by the definition-specific standard deviation of the within-branch exposure change, calculated in the common manager-choice sample. Panel A contains 1,062 branch-wave observations from 531 branches; Panel B contains 644 observations from 322 balanced branches. The explicit fixed-2018 construction reproduces the stored headline exposure exactly.

C.2 Manager-ethnicity and delegated-lending-authority robustness

Figure C2 collects the robustness of the manager-appointment and delegation results to the sample and measurement choices described in the main text. Moving across the horizontal axis, we vary the set $\mathcal{E}_{bt}$ in equation 4 by matching conflict within 15 and then 25 kilometers of a branch of bank b rather than 20; we exclude the state-owned banks, CBE and CBB ("Exclude public"); we exclude branches in Addis Ababa ("No Addis"); we drop branches whose manager ethnicity is imputed rather than observed ("No imputation", see Section I); and we drop branches in Tigray ("No Tigray"). Panel A reports the three manager-ethnicity outcomes and Panel B the delegated lending authority. The estimates are stable across every variant, both quantitatively and qualitatively: none of these choices drives the results. Table C1 reports the remaining check, dropping the branches of banks of each ethnic group one at a time, which does not fit the common scale of the figure.

Figure C2: Manager ethnicity and delegated lending authority: selected robustness checks



Notes: The independent variable is log bank-network conflict exposure involving the bank's ethnic group and excluding conflict near the focal branch; markers give effects of a one-standard-deviation increase. Regressions include branch and period fixed effects in non-bank-coethnic cities; whiskers are 95% confidence intervals using branch-clustered standard errors. Panel A reports manager-ethnicity outcomes in percentage points (531 baseline branches). Panel B reports the maximum loan a manager may approve without headquarters' approval, in millions of 2018 ETB (322 balanced baseline branches). "Exclude public" drops CBE and CBB; "No Tigray" drops branches, not Tigray-related events. The Addis and non-Addis delegation slopes are not statistically distinguishable with group-specific period effects ($p = 0.80$).

Table C1: Robustness: dropping individual ethnic groups

	Panel A: Branches in Bank-Coethnic Cities							
	City coethnic manager				Bank coethnic manager			
	Drop Oromo	Drop Tigray	Drop Amhara	Drop SNNP	Drop Oromo	Drop Tigray	Drop Amhara	Drop SNNP
Network Exposure	0.010	-0.005	-0.087	0.023	0.010	-0.005	-0.087	0.023
	(0.034)	(0.026)	(0.053)	(0.031)	(0.034)	(0.026)	(0.053)	(0.031)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.669	0.679	0.698	0.681	0.669	0.679	0.698	0.681
N (Branches)	347	448	129	420	347	448	129	420
	Panel B: Branches in Non-Bank-Coethnic Cities							
	City coethnic manager				Bank coethnic manager			
	Drop Oromo	Drop Tigray	Drop Amhara	Drop SNNP	Drop Oromo	Drop Tigray	Drop Amhara	Drop SNNP
Network Exposure	0.030*	0.079***	0.048***	0.055***	-0.062***	-0.127***	-0.097***	-0.112***
	(0.018)	(0.029)	(0.016)	(0.018)	(0.016)	(0.031)	(0.015)	(0.017)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.178	0.197	0.222	0.156	0.775	0.755	0.735	0.803
p-value	0.605	0.031	0.014	0.377	0.056	0.003	0.857	0.000
N (Branches)	404	458	275	456	404	458	275	456

Notes: The dependent variable is an indicator that equals 1 when a branch's manager is coethnic with the ethnic group associated with the city (columns 1–4) or with the ethnic group associated with the branch's bank (columns 5–8). Each column drops all branches belonging to banks of the indicated ethnic group. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the test that the slope on Network Exposure is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.3 The effect of conflict on employees being local

Table C2: Bank-level exposure to ethnic conflict and local staff

	Panel A: Branches in Bank-Coethnic Cities					
	Manager born in region			Share local employees		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-0.037	-0.107	-0.037	-0.027*	0.013	-0.027**
	(0.030)	(0.094)	(0.030)	(0.014)	(0.051)	(0.014)
Local Exposure			0.008			0.012**
			(0.012)			(0.005)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.421	0.421	0.421	0.556	0.556	0.556
N (Branches)	214	214	214	448	448	448
	Panel B: Branches in Non-Bank-Coethnic Cities					
	Manager born in region			Share local employees		
Network Exposure	-0.025	-0.026	-0.024	-0.008	-0.018*	-0.009
	(0.027)	(0.028)	(0.027)	(0.010)	(0.010)	(0.010)
Local Exposure			-0.006			0.020***
			(0.018)			(0.005)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.448	0.448	0.448	0.584	0.584	0.584
p-value	0.762	0.413	0.741	0.265	0.551	0.279
N (Branches)	250	250	250	531	531	531

Notes: This table shows the results of a regression on two measures of whether staff comes from the same region as the branch. Share local employees is the share of employees who were born in the same city as the branch, local manager is a dummy for whether the manager was born in the same region as the branch. Note that this question was added to the survey after around 500 branches were interviewed in 2018 resulting in a small sample size. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 5. The mean (SD) of the within-branch change in log Network Exposure from 2018 to 2022 is 0.955 (1.340). Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the test that the slope on Network Exposure is identical in bank-coethnic and non bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.4 Alternative transformation of the conflict variables

In this section, we report the results from the specification in equation 6 using an asinh transformation ($\text{asinh}(x) = \ln\left(x + \sqrt{x^2+1}\right)$) of the network exposure variable

Table C3: IHS Robustness: Branch Managers

	(1) City coethnic manager	(2) Bank coethnic manager	(3) Delegation	(4) Tenure in bank	(5) Tenure in branch	(6) Age
Panel A: Bank-City Coethnic						
IHS Network Exposure	-0.009	-0.009	-120.900	0.051	-0.340**	-0.033
	(0.027)	(0.027)	(474.728)	(0.249)	(0.140)	(0.386)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.680	0.680	1852.814	6.859	2.390	33.825
N	447	447	231	447	446	417
Panel B: Not Bank-City Coethnic						
	City coethnic manager	Bank coethnic manager	Delegation	Tenure in bank	Tenure in branch	Age
IHS Network Exposure	0.055***	-0.105***	-689.922**	0.538***	0.231**	0.586***
	(0.017)	(0.016)	(312.323)	(0.167)	(0.103)	(0.220)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.185	0.770	1684.783	6.837	2.366	33.700
p-value	0.050	0.003	0.317	0.105	0.001	0.164
N	531	531	322	528	528	500

Notes: Each column reports a separate regression using the inverse hyperbolic sine transformation of Network Exposure in place of the log transformation used in the main text. The delegation outcome is the largest loan a branch manager can approve without headquarters' approval, expressed in 2018 prices and winsorized at the 95th percentile within period-by-bank-city-coethnicity cells. Panel A restricts the sample to branches in coethnic cities, while Panel B focuses on branches in non-coethnic cities. All specifications include branch and time fixed effects. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the t-test that the parameter estimates for branches in bank-coethnic cities (Panel A) and in non bank-coethnic cities (Panel B) are the same. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

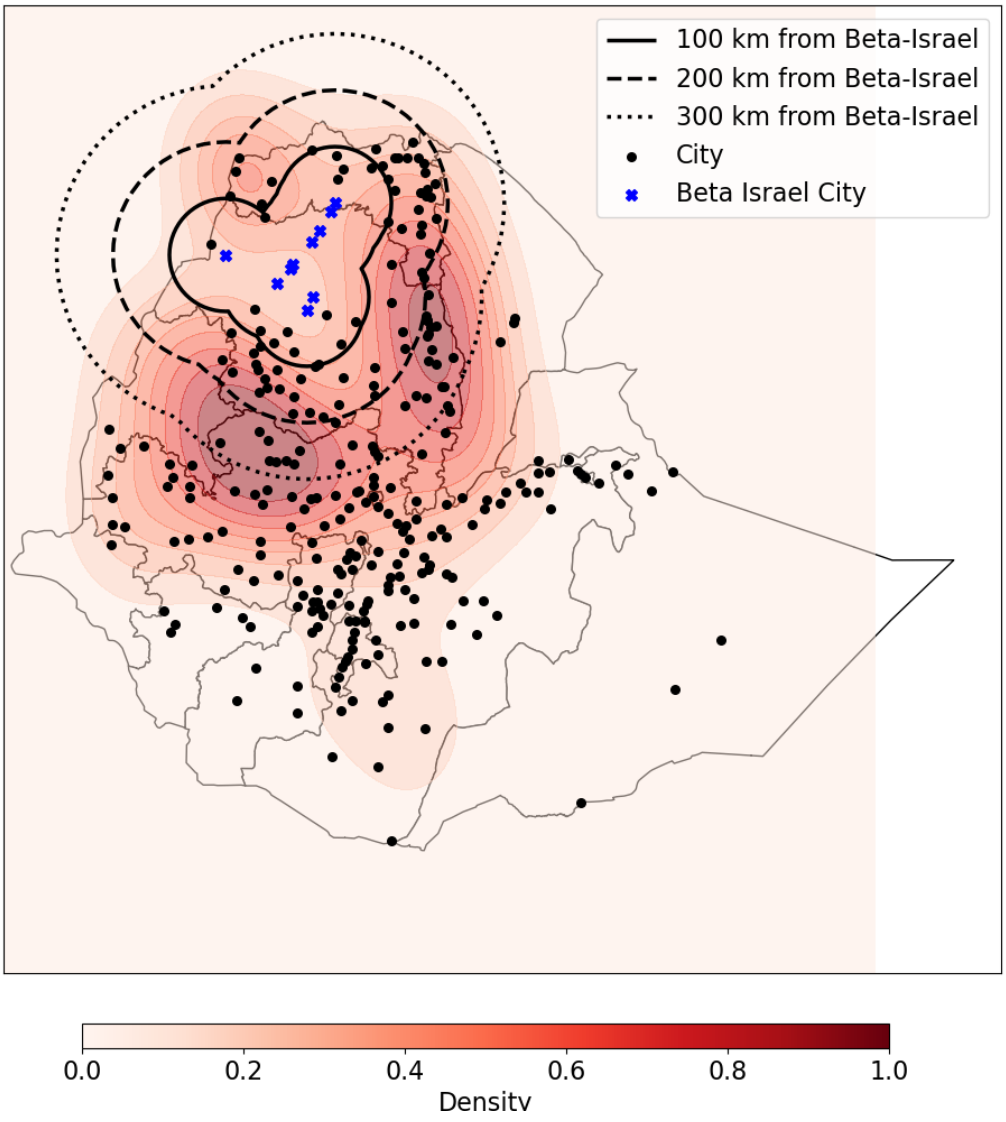
Table C4: IHS Robustness: Lending Activities

	Panel A: Bank-City Coethnic					
	(1) Oper. distance	(2) Num. loans	(3) Loan size	(4) Lending rate	(5) Collateral	(6) Default
IHS Network Exposure	-2.016***	0.074	0.527***	0.002	0.016	-0.027
	(0.610)	(0.120)	(0.159)	(0.001)	(0.014)	(0.020)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.818	2.181	6.041	0.147	0.986	0.052
N	308	224	222	292	202	273
	Panel B: Not Bank-City Coethnic					
	Oper. distance	Num. loans	Loan size	Lending rate	Collateral	Default
IHS Network Exposure	-0.073	-0.248***	0.151	-0.000	-0.006	-0.008
	(0.168)	(0.056)	(0.144)	(0.001)	(0.009)	(0.008)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.542	1.928	6.255	0.148	0.995	0.045
p-value	0.002	0.015	0.080	0.337	0.196	0.359
N	413	307	304	366	267	325

Notes: Each column reports a separate regression using the inverse hyperbolic sine transformation of Network Exposure in place of the log transformation used in the main text. The dependent variables are the operational distance of the branch, the log of the number of loans outstanding, the log of the average loan size (deflated to 2018 prices), the lending rate on a typical loan, the percentage of collateral on a typical loan, and the share of loans that default. Panel A restricts the sample to branches in coethnic cities, while Panel B focuses on branches in non-coethnic cities. All specifications include branch and time fixed effects. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the t-test that the parameter estimates for branches in bank-coethnic cities (Panel A) and in non bank-coethnic cities (Panel B) are the same. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.5 Beta Israel cities and conflict geography

Figure C3: Beta Israel cities and conflict contours



Notes: Blue crosses mark cities historically associated with Beta Israel communities, and black dots mark other cities. The solid, dashed, and dotted black contours indicate locations within 100, 200, and 300 km, respectively, of the nearest historical Beta Israel city. The red shading displays the density of fatalities from ethnic conflict between 2015 and 2022, with darker areas indicating greater density.

C.6 Beta-Israel exposure and lending

Table C5 reports the lending results using the preferred 100 km Beta-Israel exposure measure. The pattern is similar to that in the baseline specification, although it highlights greater adjustment to loan terms in bank-coethnic cities. In bank-coethnic cities, exposure to Beta-Israel cities is associated with larger average loan sizes, higher collateral requirements, and lower default rates, consistent with a shift toward safer, collateral-backed lending. In non-bank-coethnic cities, banks expand their operational radius and provide fewer, larger loans. The estimates suggest that banks can continue lending effectively outside cities where the majority of customers share the bank's ethnicity.

Table C5: Bank-level exposure to Beta-Israel cities and lending (100 km)

	Panel A: Bank and city coethnic					
	(1) Oper. distance	(2) Num. loans	(3) Loan size	(4) Lending rate	(5) Collateral	(6) Default
Beta Israel Exposure	-4.642***	0.040	1.512***	0.000	0.232***	-0.128***
	(1.501)	(0.289)	(0.531)	(0.004)	(0.047)	(0.048)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.818	2.181	6.041	0.147	0.986	0.052
N	308	224	222	292	202	273
	Panel B: Bank and city not coethnic					
	Oper. distance	Num. loans	Loan size	Lending rate	Collateral	Default
Beta Israel Exposure	7.116***	-0.768**	3.053***	0.002	-0.084	0.056
	(1.284)	(0.364)	(0.772)	(0.006)	(0.052)	(0.059)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.542	1.928	6.255	0.148	0.995	0.045
p-value	0.000	0.083	0.101	0.749	0.000	0.016
N	413	307	304	366	267	325

Notes: The dependent variables are the operational distance of the branch, the log number of outstanding loans, the log average loan size, the lending rate on a typical loan, the percentage of collateral on a typical loan, and the share of loans in default. Beta Israel Exposure is the share of each bank's 2018 branches within 100 km of a historic Beta-Israel city, normalized to lie in $[0, 1]$ and interacted with a post-2018 indicator. Panel A restricts the sample to branches in bank-coethnic cities; Panel B focuses on branches in non-bank-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value tests whether the coefficient is identical across panels. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.7 Robustness: 50 km Beta-Israel exposure window

Section 4.5 measures bank-level Beta-Israel exposure as the share of a bank's branches within 100 km of a historic Beta-Israel city. We use 100 km because the contested 1992 borders that link the Beta-Israel migration to subsequent ethnic conflict extend that far from the cities themselves. To check whether the radius drives our results, we re-estimate Table 8 and Table C5 using a tighter 50 km window. We construct $\texttt{share_branch_beta_50}_b$ analogously and apply the same normalization.

Table C6 reports the manager results. The patterns mirror the 100 km benchmark. In non bank-coethnic cities, exposure raises the share of city-coethnic managers and lowers the share of bank-coethnic managers; these effects, together with the reduction in delegation, all remain significant at the 1% level. The point estimates are 75–80% of the 100 km magnitudes, since the tighter radius sets many branches' exposure to zero and so removes variation. The p-value for test of a different effect in bank coethnic cities for the city-coethnic outcome rises from 0.029 to 0.052, while the bank-coethnic interaction stays highly significant ($p = 0.003$).

Table C7 reports the lending results. Bank-coethnic and non bank-coethnic markets again diverge: in coethnic cities, exposure shifts lending toward larger, more collateralized, lower-default loans; in non-coethnic cities, branches serve customers over a wider area and grant larger loans. Every coefficient retains its sign, and all but two retain their significance level. We keep 100 km as the preferred specification because it more closely matches the geographic span of the contested 1992 borders.

Table C8 tightens the window further, reporting the in-city definition – the share of a bank's 2018 branches located in a historic Beta Israel city – alongside the 50 km definition for the same set of outcomes. No other sampled city lies within 20 km of a Beta Israel city, so the in-city measure is also the 20 km measure in this sample. The manager-appointment and delegation results survive at the tightest definition available.

Table C6: Bank-level exposure to Beta-Israel cities and branch managers (50 km)

	Panel A: Bank-City Coethnic					
	(1) City coethnic manager	(2) Bank coethnic manager	(3) Coethnic with neither	(4) Delegation	(5) Tenure in bank	(6) Tenure in branch
Beta Israel Exposure	0.020	0.020	-0.020	-1498	-1.781*	-1.319***
	(0.090)	(0.090)	(0.090)	(1944)	(0.998)	(0.448)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.680	0.680	0.320	1852.8	6.859	2.390
N	447	447	447	231	447	446
	Panel B: Not Bank-City Coethnic					
	City coethnic manager	Bank coethnic manager	Coethnic with neither	Delegation	Tenure in bank	Tenure in branch
Beta Israel Exposure	0.287***	-0.403***	0.116*	-5301***	0.121	-0.149
	(0.089)	(0.092)	(0.061)	(1460)	(0.872)	(0.683)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.185	0.770	0.045	1684.8	6.837	2.366
p-value	0.035	0.001	0.213	0.118	0.152	0.152
N	531	531	531	322	528	528

Notes: Replication of Table 8 using a 50 km radius around historic Beta-Israel cities. The dependent variables are indicators for a city-coethnic manager (column 1), a bank-coethnic manager (column 2), and a manager coethnic with neither group (column 3); the largest loan the manager can approve without headquarters' agreement (column 4), measured in thousands of 2018 ETB and winsorized at the 95th percentile within period $\times$ bank-city-coethnicity cells; and tenure in the bank (column 5) and branch (column 6), in years. In non-bank-coethnic cities, the three ethnicity categories in columns 1–3 are mutually exclusive and exhaustive, so their coefficients sum to zero by construction. Beta Israel Exposure is the normalized share of each bank's 2018 branch network located within 50 km of a historic Beta-Israel city, interacted with a post-2018 indicator. Panel A restricts the sample to branches in bank-coethnic cities; Panel B focuses on branches in non-bank-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value tests whether the coefficient is identical across panels. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table C7: Bank-level exposure to Beta-Israel cities and lending (50 km)

	Panel A: Bank-City Coethnic					
	(1) Oper. distance	(2) Num. loans	(3) Loan size	(4) Lending rate	(5) Collateral	(6) Default
Beta Israel Exposure	-3.428**	0.128	3.525***	0.005	0.194***	-0.118***
	(1.488)	(0.319)	(0.845)	(0.005)	(0.054)	(0.043)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.818	2.181	6.041	0.147	0.986	0.052
N	308	224	222	292	202	273
	Panel B: Not Bank-City Coethnic					
	Oper. distance	Num. loans	Loan size	Lending rate	Collateral	Default
Beta Israel Exposure	8.100***	-0.292	3.931***	0.006	-0.091**	0.034
	(1.254)	(0.325)	(0.886)	(0.005)	(0.043)	(0.045)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.542	1.928	6.255	0.148	0.995	0.045
p-value	0.000	0.357	0.741	0.862	0.000	0.015
N	413	307	304	366	267	325

Notes: Replication of Table C5 using a 50 km radius around historic Beta-Israel cities to construct the bank-level exposure measure. The dependent variables are the operational distance of the branch, the log number of outstanding loans, the log average loan size, the lending rate on a typical loan, the percentage of collateral on a typical loan, and the share of loans in default. Beta Israel Exposure is the share of each bank's branches within 50 km of a historic Beta-Israel city, normalized to lie in $[0, 1]$ and interacted with a post-2018 dummy. Panel A restricts the sample to branches in coethnic cities; Panel B focuses on branches in non-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the t-test that the coefficient is identical across panels. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table C8: Bank-level exposure to Beta-Israel cities and branch managers (in-city and 50 km)

	Bank and city non-coethnic		Bank and city coethnic	
Outcome	In-city	50 km	In-city	50 km
City-coethnic manager	0.297***	0.287***	0.020	0.020
	(0.100)	(0.089)	(0.106)	(0.090)
Bank-coethnic manager	-0.425***	-0.403***	0.020	0.020
	(0.102)	(0.092)	(0.106)	(0.090)
Neither-coethnic manager	0.129*	0.116*	-0.020	-0.020
	(0.072)	(0.061)	(0.106)	(0.090)
Delegation	-6005***	-5301***	-2174	-1498
	(1610)	(1460)	(2277)	(1944)
Tenure in bank	0.901	0.121	-0.736	-1.781*
	(1.018)	(0.872)	(1.239)	(0.998)
Tenure in branch	-0.209	-0.149	-1.325**	-1.319***
	(0.751)	(0.683)	(0.514)	(0.448)

Notes: In-city exposure is the share of a bank's 2018 branches located in a historical Beta Israel city; 50 km exposure is the share of branches in or within 50 km of one. No other sampled city lies within 20 km, so the in-city measure is also the 20 km measure in this sample. Each period-2 share is divided by its own range, and period-1 exposure is set to zero. Regressions are estimated separately by bank–city coethnicity and include branch and time fixed effects. Parentheses contain branch-clustered standard errors. Delegation is in thousands of 2018 ETB. Stars denote 10, 5, and 1 percent significance.

D Wild Cluster Bootstrap Inference

Our main specifications cluster standard errors at the branch level, reflecting the panel structure of the data. However, network exposure — the key independent variable — varies at the bank level, with only 16 banks in the sample. If bank-level shocks induce residual correlation across branches of the same bank, branch-level clustering may overstate statistical precision. Standard asymptotic corrections for cluster-robust standard errors perform poorly with so few clusters (A Colin Cameron, Jonah B Gelbach and Douglas L Miller, 2008).

To address this concern, we re-estimate all specifications using wild cluster bootstrap inference at the bank level. We use the `boottest` package (David Roodman, Morten Ørregaard Nielsen, James G MacKinnon and Matthew D Webb, 2019) with Webb weights and 9,999 bootstrap replications. Each table below presents two panels: Panel A reports results for branches in bank-coethnic cities (BCC), and Panel B for branches in non-bank-coethnic cities (NBCC). Both panels report the original point estimates and branch-clustered standard errors in parentheses, with bank-level bootstrap p-values in square brackets. In Panel B, the last row additionally reports the bootstrap p-value for the test that the effect of conflict differs between the two subsamples (i.e., the interaction test from equation 7). The paper's central prediction concerns Panel B; Panel A serves as a comparison.

Table D1 reports results for the manager ethnicity regressions. All six network-exposure coefficients in non-bank-coethnic cities remain significant at the 5% level under bank-level clustering, with bootstrap p-values between 0.014 and 0.027. The differential effect between the two subsamples is significant at the 5% level or better for three of the six specifications.

Table D2 reports results for delegation. The point estimates are negative and significant within non-bank-coethnic branches using the original branch-clustered standard errors. Under bank-level wild-bootstrap inference, the corresponding p-values are 0.088, 0.168, and 0.091: the baseline and local-conflict estimates are marginally significant at 10 percent, while the city-by-time estimate is not. The bootstrap p-values for the differential relative to bank-coethnic branches are 0.459, 0.713, and 0.464.

Tables D3–D4 report results for manager reallocation and characteristics. All four reallocation outcomes are marginally significant at the 10% level. Manager tenure in the branch remains significant and tenure in the bank is marginally significant, while manager age is not. The differential effect on tenure in the branch remains significant; that on tenure in the bank does not.

Table D5 reports results for lending outcomes. Most lending outcomes lose significance, with the exception of the log number of loans ($p = 0.017$), which remains significant at the 5% level. This reflects the limited statistical power of 16 bank-level clusters for detecting effects on outcomes that are plausibly noisier than the manager ethnicity outcome. The point estimates and branch-clustered standard errors are unchanged; the loss of significance reflects the more conservative inference, not a change in the estimated effects.

In summary, bank-level bootstrap inference preserves the manager-selection result and the effect on the number of loans. Delegation is marginally significant in the baseline and local-conflict specifications, imprecise with city-by-time effects, and not statistically different across the two city groups.

Table D1: Wild cluster bootstrap inference: manager ethnicity

	Panel A: Branches in Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-0.005	0.070	-0.006	-0.005	0.070	-0.006
	(0.026)	(0.055)	(0.026)	(0.026)	(0.055)	(0.026)
	[0.886]	[0.323]	[0.863]	[0.886]	[0.323]	[0.863]
Local Exposure			0.014			0.014
			(0.008)			(0.008)
			[.]			[.]
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.680	0.695	0.680	0.680	0.695	0.680
N (Branches)	447	371	447	447	371	447

	Panel B: Branches in Non-Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	0.050	0.047	0.051	-0.097	-0.096	-0.099
	(0.016)	(0.017)	(0.016)	(0.015)	(0.017)	(0.015)
	[0.025]	[0.019]	[0.027]	[0.014]	[0.021]	[0.015]
Local Exposure			-0.014			0.031
			(0.009)			(0.010)
			[.]			[.]
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.185	0.181	0.185	0.770	0.769	0.770
p-value diff vs BCC (bootstrap)	0.258	0.834	0.252	0.006	0.050	0.001
N (Branches)	531	464	531	531	464	531

Notes: This table replicates Table 2. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap p-values for Network Exposure are reported in square brackets, computed using Webb weights with 9,999 replications and clustering at the bank level (16 banks); the bootstrap p-value for the local-exposure control is not reported. The last row of Panel B reports the bootstrap p-value for the null that the effect of network exposure is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D2: Wild cluster bootstrap inference: delegation

	Panel A: Branches in Bank-Coethnic Cities		
	(1)	(2)	(3)
Network Exposure	-111.9	-1178.3	-115.6
	(381.9)	(839.1)	(384.5)
	[0.861]	[0.406]	[0.855]
Local Exposure			-422.3
			(172.1)
			[.]
Branch FE	Yes	Yes	Yes
Time FE	Yes	No	Yes
City-Time FE	No	Yes	No
2018 Mean dep. var	1852.8	1993.8	1852.8
N (Branches)	231	160	231

	Panel B: Branches in Non-Bank-Coethnic Cities		
	(1)	(2)	(3)
Network Exposure	-633.1	-666.1	-627.0
	(282.0)	(330.3)	(281.5)
	[0.088]	[0.168]	[0.091]
Local Exposure			-60.7
			(135.6)
			[.]
Branch FE	Yes	Yes	Yes
Time FE	Yes	No	Yes
City-Time FE	No	Yes	No
2018 Mean dep. var	1684.8	1721.8	1684.8
p-value diff vs BCC (bootstrap)	0.459	0.713	0.464
N (Branches)	322	266	322

Notes: This table replicates Table 3. The dependent variable is the largest loan a branch manager can approve without headquarters' agreement, measured in thousands of 2018 ETB and winsorized at the 95th percentile. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap p-values for Network Exposure are reported in square brackets; the bootstrap p-value for the local-exposure control is not reported. The last row of Panel B reports the bootstrap p-value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D3: Wild cluster bootstrap inference: manager reallocation

	Panel A: Branches in Bank-Coethnic Cities			
	New manager bank	New manager branch	Reallocated	New hire
	(1)	(2)	(3)	(4)
Network Exposure	0.016	-0.015	-0.027	0.011
	(0.009)	(0.032)	(0.031)	(0.008)
	[0.234]	[0.618]	[0.278]	[0.307]
Mean dep. var	0.034	0.397	0.370	0.027
N (Branches)	447	446	446	446
	Panel B: Branches in Non-Bank-Coethnic Cities			
	New manager bank	New manager branch	Reallocated	New hire
	(1)	(2)	(3)	(4)
Network Exposure	-0.010	0.065	0.075	-0.010
	(0.008)	(0.024)	(0.024)	(0.008)
	[0.059]	[0.083]	[0.084]	[0.059]
Mean dep. var	0.017	0.360	0.345	0.015
p-value diff vs BCC (bootstrap)	0.141	0.387	0.094	0.151
N (Branches)	528	528	528	528

Notes: This table replicates Table 4. The sample is restricted to the 2022 cross-section. The four outcomes are whether the manager joined the bank (column 1) or branch (column 2) after the onset of conflict, and conditional on joining the branch, whether they were reallocated (column 3) or a new hire (column 4). Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap p-values are reported in square brackets. The last row of Panel B reports the bootstrap p-value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D4: Wild cluster bootstrap inference: manager characteristics

	Panel A: Branches in Bank-Coethnic Cities				
	Tenure in bank	Tenure in branch	Age	University degree	Male manager
	(1)	(2)	(3)	(4)	(5)
Network Exposure	0.029	-0.265	-0.026	-0.002	0.008
	(0.230)	(0.126)	(0.354)	(0.002)	(0.028)
	[0.907]	[0.019]	[0.957]	[0.444]	[0.897]
Branch FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	6.859	2.390	33.825	0.993	0.868
N (Branches)	447	446	417	448	448
	Panel B: Branches in Non-Bank-Coethnic Cities				
	Tenure in bank	Tenure in branch	Age	University degree	Male manager
	(1)	(2)	(3)	(4)	(5)
Network Exposure	0.498	0.221	0.538	-0.002	0.008
	(0.154)	(0.098)	(0.202)	(0.001)	(0.014)
	[0.074]	[0.018]	[0.116]	[0.497]	[0.560]
Branch FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	6.837	2.366	33.700	0.989	0.866
p-value diff vs BCC (bootstrap)	0.116	0.004	0.257	0.952	0.998
N (Branches)	528	528	500	529	529

Notes: This table replicates Table 5. The dependent variables are the manager's tenure in the bank, tenure in the current branch, age, whether they hold a university degree, and gender. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap p-values are reported in square brackets. The last row of Panel B reports the bootstrap p-value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D5: Wild cluster bootstrap inference: lending outcomes

	Panel A: Branches in Bank-Coethnic Cities					
	Operational distance	Log number of loans	Log average loan size	Average lending rate	Collateral percentage	Share loan default
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-1.658	0.049	0.395	0.001	0.015	-0.026
	(0.681)	(0.117)	(0.158)	(0.001)	(0.013)	(0.017)
	[0.006]	[0.591]	[0.244]	[0.528]	[0.663]	[0.136]
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.818	2.181	6.041	0.147	0.986	0.052
N (Branches)	308	224	222	292	202	273

	Panel B: Branches in Non-Bank-Coethnic Cities					
	Operational distance	Log number of loans	Log average loan size	Average lending rate	Collateral percentage	Share loan default
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-0.084	-0.233	0.123	-0.000	-0.006	-0.007
	(0.157)	(0.051)	(0.133)	(0.001)	(0.008)	(0.007)
	[0.832]	[0.017]	[0.584]	[0.958]	[0.526]	[0.444]
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.542	1.928	6.255	0.148	0.995	0.045
p-value diff vs BCC (bootstrap)	0.018	0.001	0.488	0.638	0.512	0.152
N (Branches)	413	307	304	366	267	325

Notes: This table replicates Table 6. The dependent variables are the operational distance, log number of loans, log average loan size, average lending rate, collateral percentage, and share of loans in default. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap p-values are reported in square brackets. The last row of Panel B reports the bootstrap p-value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E Ethnicity of banks, cities and managers

E.1 Comments on specific choices made in the analysis

Bank ethnicity. We classify each of the 16 commercial banks in our sample into a single dominant ethnicity using information from the 2018 annual reports. For each bank, we collect the names of all members of the board of directors and identify each director's ethnic group from their name, following the conventional correspondence between Ethiopian naming conventions and ethnic affiliation. A bank is assigned the majority ethnicity of its board. The two state-owned banks, the Commercial Bank of Ethiopia and the Construction and Business Bank, are classified as Amhara; the latter was effectively absorbed into the former during our sample period. This procedure yields 9 Amhara-dominated banks, 3 Oromo, 2 SNNP, and 2 Tigray.

City ethnicity. We assign each city its historical ethnic homeland using the digitized ethnic-boundary shapefile from Nathan Nunn (2008), which encodes the ethnic territories of George Peter Murdock (1959). We spatially match each branch's geographic coordinates to a Murdock polygon using the `geoinpoly` routine. Murdock's tribe labels are pre-modern ethnographic categories (e.g., "Galla," "Somal"); we map these to the contemporary ethnic categories used in our analysis (Oromo, Amhara, Tigray, Somali, SNNP, Beta Israel, Other) using a correspondence table constructed from Wikipedia entries for each Murdock tribe. The full mapping is available in the replication package (`1rawData/13EthnicGroups/tribeEthnicityMapping.xlsx`). Addis Ababa, which lies on the boundary between Oromo and Amhara territories in Murdock's classification, is assigned to the Amhara category, reflecting the dominant ethnic composition of the city's banking sector and civil service during our sample period.

Treatment of the Harari. The Harari are a small ethnic group (approximately 0.2% of the Ethiopian population in the 2007 census) whose traditional homeland is the walled city of Harar and its immediate surroundings, located within predominantly Oromo territory. No manager in our survey sample self-identifies as Harari, so Harari does not appear as a category in the manager-ethnicity classification. For the city-ethnicity assignment based on the Nunn (2008) digitization of the Murdock Atlas, Harar and its surrounding cities fall within the Oromo polygon and are therefore assigned to the Oromo category. This coding has no material effect on our results: only 7 branches in our panel are located in cities of the historic Harari homeland.

Aggregation of SNNP groups. The Southern Nations, Nationalities, and Peoples (SNNP) region of Ethiopia is home to more than 50 officially recognized ethnic groups, eight of which appear in our sample: Sidama, Gurage, Gamo, Wolayta, Hadiya, Kambaata, Kafa, and Silte. Because bank and city ethnicity in our main specifications are defined at a coarser level (the bank's dominant ethnicity and the city's Murdock-Atlas classification), we aggregate these eight groups into a single "SNNP" category for the coethnicity indicators. This follows standard practice in the Ethiopian ethnic-politics literature and is also consistent with the self-identification reported by several managers in our data.

E.2 List of banks and bank-ethnicities

Table E1: Distribution of banks by ethnic affiliation

Bank name	Bank ethnicity	Count	Share (%)
Abay	Amhara	80	8.2
Addis	Amhara	5	0.5
Awash	Oromo	100	10.2
BOA	Amhara	37	3.8
Berhan	Amhara	72	7.4
Bunna International	Amhara	46	4.7
CBB	Amhara	19	1.9
CBE	Amhara	210	21.5
CBO	Oromo	53	5.4
Dashen	Amhara	55	5.6
Debub	SNNP	17	1.7
Lion	Tigray	14	1.4
NIB International	SNNP	86	8.8
OIB	Oromo	75	7.7
United	Amhara	51	5.2
Wegagen	Tigray	59	6.0

Notes: This table reports the ethnic affiliation of each bank in the sample, the number of surveyed branches, and the corresponding share of the total sample. Bank ethnicity is the dominant ethnic affiliation of the bank's leadership.

E.3 Number of cities by ethnic group

Table E2: Distribution of cities by ethnic group

City ethnicity	Count	Share (%)
Afar	4	0.4
Amhara	522	53.3
Anuak	4	0.4
Benishangul-Gumuz	2	0.2
Oromo	260	26.6
SNNP	160	16.3
Somali	17	1.7
Tigray	10	1.0

Notes: This table reports the distribution of branch locations by ethnic group. City ethnicity is determined using the Murdock Ethnographic Atlas. Count and share refer to the number and percentage of branches located in cities of each ethnic group.

E.4 Number of managers by ethnic group by period

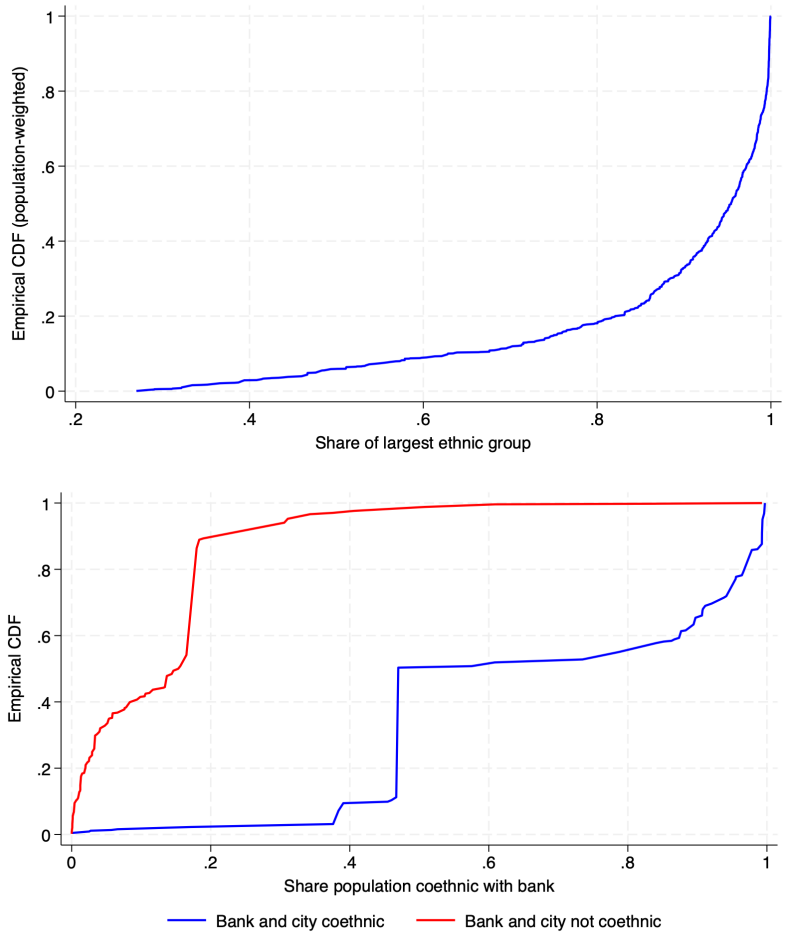
Table E3: Distribution of branch managers' ethnicity by period

Manager ethnicity	First Wave (2018)		Second Wave (2022)	
	Count	Share (%)	Count	Share (%)
Afar	1	0.1	5	0.5
Agew	0	0.0	8	0.8
Amhara	495	50.6	433	44.2
Benishangul-Gumuz	2	0.2	1	0.1
Oromo	289	29.5	380	38.8
SNNP	109	11.1	112	11.4
Somali	2	0.2	19	1.9
Tigray	81	8.3	20	2.0

Notes: This table reports the distribution of branch managers' ethnicity across survey waves. Manager ethnicity is identified using multiple indicators including region of birth, accent, spoken languages, and name, determined jointly by two interviewers.

E.5 Ethnic diversity at the local level

Figure E1: Ethnic diversity in Ethiopia



Notes: This figure shows, in the top panel, the empirical CDF for the share of the population that belongs to largest ethnic group in each woreda – the smallest administrative unit – in Ethiopia. It shows that in over 80% of woredas over 80% of people belong to a single ethnicity. The bottom panel instead shows that share of the population in each zone (region in the case of Addis Ababa) in which a branch is located that belongs to the same ethnic group of the bank. The differences in administrative units used in this figure is based on how (im)precisely we observe the location of bank branches. This is based on 2007 IPUMS data and corresponding administrative boundaries.

F Management Practices Survey and Methodology

F.1 Scoring Management Performances

The concept of "good management practice" is often relative and contingent to a specific market or sector. However, international best practices in the banking sector have been widely studied, allowing the literature to reach a consensus on what constitutes a "good" practice. We evaluate and score management practices by defining the concept of "good" and "bad" practice and codifying it from 1 (worst practice) to 5 (best practice) across eight key dimensions. These practices are grouped into four areas: lean management, performance management, target management, and people management.[1]

The lean management section evaluates the efficiency of day-to-day operations (e.g., organization of the workflow, management of slack time, method of staff assignment to specific tasks, etc.) and assesses whether the branch has adopted recognized best practices. The performance management part measures how performance is tracked and reviewed (e.g., what KPIs are used, who oversees the performance review, how frequently the performance measures are revised, etc.). The target management section reviews the nature and scope of a branch's targets (e.g., the relative importance of financial targets compared to non-financial ones, how these targets are linked to performance measures, etc.) and assesses whether they align with the bank's objectives. Finally, the people management part tests whether good performance is appropriately rewarded and whether bad performance is sanctioned (e.g., adoption of an articulated appraisal system, use of a reward plan, presence of sanctions for underperformers, etc.).

An overview of these four sections is presented in Table F1. After the completion of the survey, we combine these scores to obtain a unique index of management practices. Since the scaling may vary across practices in the econometric estimation, we normalize the final scores to z-scores (i.e., mean zero and standard deviation one). To build our final management index, we take the unweighted average across all z-scores as

[1] Note, this section is taken directly from a previously circulated working paper (Nicola Gennaioli, Nicola Limodio and Francesco Strobbe, 2019)

our primary measure of overall management practice.

Table F1: Survey Indicators

(1) Indicator	(2) Objective	(3) Measure
	1 Lean Management	
1.1	Introduction of new management techniques	Tests whether operational efficiencies have been introduced and why
1.2	Good use of human resources	Tests whether incoming demand is segmented appropriately and the efficiency at matching supply and demand of skills
	2 Performance Management	
2.1	Performance tracking	Tests whether performance is tracked using meaningful metrics and with appropriate regularity
2.2	Review of performance	Tests whether performance is reviewed with appropriate frequency and communicated with staff
	3 Target Management	
3.1	Target balance	Tests whether targets cover a sufficiently broad set of metrics
3.2	Target interconnections	Tests whether targets are tied to company objectives and how well they cascade down the organization
	4 People Management	
4.1	Building a high performance culture	Tests whether good performance is rewarded proportionately
4.2	Making room for talent	Tests whether the branch is able to deal with underperformers

Notes: This table illustrates our survey's section on management performance. To measure good managerial practices, we evaluate them along four main categories, i.e., lean management, performance management, target management, and people management. Each category is scored on two dimensions on a scale from 1 (bad practice) to 5 (good practice). Our final management performance index is calculated by standardizing each of the eight resulting scores and by taking their unweighted average. Each of the sub-indices is calculated in the same way using a combination of two metrics (respectively, the lean, performance, target and people management practices).

F.2 Responses Collection

The information content and precision of our management performance index crucially depend on the quality of the responses collected through our survey. For this reason, we structured our questions to allow respondents to provide accurate and unbiased answers. As pointed out by (Marianne Bertrand and Sendhil Mullainathan, 2001), responses are typically biased by the presence of a scoring scale, which may prompt respondents to provide the interviewer's expected answer. Additionally, interviewers may have preconceptions toward the interviewed managers. Apart from these issues, many background factors correlated with management behavior may systematically bias survey data.

To counter these potential negative effects, we adopt a combination of strategies tested in the existing literature. First, we ensure accurate responses by conducting telephone surveys without informing managers that their answers will be evaluated against a scoring grid, allowing us to gather information about actual management practices. Second, we ask a series of open-ended questions (e.g., "What types of targets are set for the bank in general? What are the goals for your branch?") and record responses until an accurate assessment of management practices is possible. Third, to correct any inconsistent interpretation of responses, we ensure interviewers conduct a minimum number of training interviews during a pilot survey phase. Furthermore, since almost all our interviewers conducted over 40 interviews (with an average of 200 interviews per interviewer), we are able to control for interviewer fixed effects in a robustness section. Fourth, we adopt a double-scoring technique, where another interviewer silently listens and scores the responses provided during the interview. Finally, we collect information on a large set of manager characteristics (e.g., age, education, ethnicity, etc.) to control for potential confounders.

F.3 Manager Participation

The average duration of an interview was approximately 27 minutes. Interviews were conducted between January 2015 and February 2018 by locally recruited research assistants (RAs). The initial survey phase consisted of a pilot involving 265 branches, which allowed us to refine the survey and train interviewers. After this stage, we surveyed the universe of Ethiopian bank branches (3,232 in total). We achieved a relatively high response rate (59%, 1,910 branches) through specific strategies. First, the survey was introduced as a neutral exercise, avoiding discussion of the bank's financial position or individual branch accounts. This ensured that interviewers were blind to financial data and helped maximize participation. Second, questions focused on evident practices within the branch, enabling branch managers to provide reliable answers. Finally, endorsement and support from the World Bank and the National Bank of Ethiopia encouraged participation by emphasizing the initiative's importance and official backing.

F.4 Additional Data Collected

In addition to management practices, our survey collected extensive data on respondent and branch characteristics. The introductory section gathered baseline information about the branch (e.g., name, geographic location, number of employees, opening date) and the manager (e.g., age, gender, education, previous employment).

A second set of questions addressed the organizational nature of the branch (e.g., management structure, autonomy from headquarters) and operations (e.g., average loan size, collateral requirements, share of

defaulted loans).

F.5 Summary statistics by wave

Tables F2 and F3 report summary statistics on the main analysis variables by survey wave for respectively organizational characteristics and characteristics related to lending activities.

Table F2: Summary statistics of organizational variables by wave

	N	Mean	Std. Dev.	Min	Max
Panel A: Manager Characteristics					
Bank coethnic manager	1,957	0.65	0.48	0	1
First Wave (2018)	979	0.73	0.45	0	1
Second Wave (2022)	978	0.57	0.49	0	1
City coethnic manager	1,957	0.54	0.50	0	1
First Wave (2018)	979	0.41	0.49	0	1
Second Wave (2022)	978	0.66	0.47	0	1
Manager born in region	1,439	0.42	0.49	0	1
First Wave (2018)	464	0.44	0.50	0	1
Second Wave (2022)	975	0.42	0.49	0	1
Tenure in bank (years)	1,954	7.71	4.44	0	36
First Wave (2018)	979	6.85	4.27	0	35
Second Wave (2022)	975	8.57	4.45	0	36
Tenure in branch (years)	1,953	2.67	2.62	0	35
First Wave (2018)	979	2.37	2.33	1	35
Second Wave (2022)	974	2.98	2.85	0	23
Age	1,896	34.00	5.67	16	60
First Wave (2018)	979	33.70	6.02	16	60
Second Wave (2022)	917	34.32	5.25	23	56
University degree	1,956	0.99	0.07	0	1
First Wave (2018)	979	0.99	0.10	0	1
Second Wave (2022)	977	1.00	0.00	1	1
Male manager	1,956	0.83	0.38	0	1
First Wave (2018)	979	0.87	0.34	0	1
Second Wave (2022)	977	0.79	0.40	0	1
Panel B: Branch Characteristics					
	N	Mean	Std. Dev.	Min	Max
Share local employees	1,920	0.57	0.25	0.10	1.00
First Wave (2018)	979	0.57	0.24	0.10	1.00
Second Wave (2022)	941	0.57	0.25	0.10	1.00
Branch employees	1,951	17.57	10.04	3	110
First Wave (2018)	979	14.77	10.32	3	110
Second Wave (2022)	972	20.40	8.90	5	78
Max loan w/o HQ	1,532	3,854.91	6,079.66	0.00	29,823.23
First Wave (2018)	979	1,681.82	1,614.30	0.00	5,500.00
Second Wave (2022)	553	7,702.04	8,642.75	149.12	29,823.23

Notes: This table reports summary statistics of organisational analysis variables by survey wave, organised into two panels. Panel A reports manager characteristics: bank and city coethnic manager indicate whether the manager shares the ethnicity of the bank or city, respectively; manager born in region indicates whether the manager was born in the same region as the branch; tenure in bank and branch are measured in years; university degree and male manager are dummy indicators. Panel B reports branch characteristics: share local employees is the share of branch employees from the local area; branch employees is the headcount; max loan w/o HQ is the largest loan a branch manager can approve without headquarters' agreement, in ETB 1,000, deflated to 2018 prices using annual CPI inflation rates (World Bank WDI, FP.CPI.TOTL.ZG) and winsorized at the 95th percentile within period $\times$ bank-city coethnicity cells.

Table F3: Summary statistics of lending variables by wave

	N	Mean	Std. Dev.	Min	Max
Operational distance (km)	1,700	6.80	4.85	0.05	50.00
First Wave (2018)	979	7.62	2.62	3.00	30.00
Second Wave (2022)	721	5.70	6.64	0.05	50.00
Log number of loans	1,510	2.43	1.34	0.00	8.52
First Wave (2018)	979	2.05	1.27	0.00	6.01
Second Wave (2022)	531	3.13	1.18	0.00	8.52
Log average loan size	1,505	6.30	3.13	-4.61	10.95
First Wave (2018)	979	6.05	3.83	-4.61	10.95
Second Wave (2022)	526	6.77	0.65	5.70	8.51
Average lending rate	1,637	0.15	0.02	0.10	0.20
First Wave (2018)	979	0.15	0.02	0.10	0.20
Second Wave (2022)	658	0.15	0.02	0.10	0.20
Collateral (share of loan)	1,448	0.95	0.19	0.50	1.50
First Wave (2018)	979	1.00	0.20	0.50	1.50
Second Wave (2022)	469	0.83	0.13	0.50	1.00
Share loan default	1,577	0.07	0.14	0.00	1.00
First Wave (2018)	979	0.05	0.09	0.00	1.00
Second Wave (2022)	598	0.09	0.20	0.00	1.00

Notes: This table reports summary statistics of lending analysis variables by survey wave. Operational distance is the maximum distance (in km) at which a branch services customers. Log number of loans and log average loan size are in natural logarithms, with loan size expressed in ETB 1,000. Average lending rate is the mean rate on the past 10 loans. Collateral is the typical collateral as a share of total loan size. Share loan default is the share of defaulting loans in the past year.

G Ethnic conflict over time and space

Figure G1: Regions of Ethiopia



Notes: This figure displays the twelve regions in Ethiopia and two administrative councils, Addis Ababa and Dire Dawa. To ensure regions stay the same over time in our analysis, we combine Central Ethiopia, South West Ethiopia, South Ethiopia and Sidama into the Southern Nations, Nationalities and Peoples' Region (SNNPR). This region was split into four between 2020 and 2023, and remains one of the most ethnically diverse regions of the country.

G.1 Mapping ACLED actors to ethnic groups

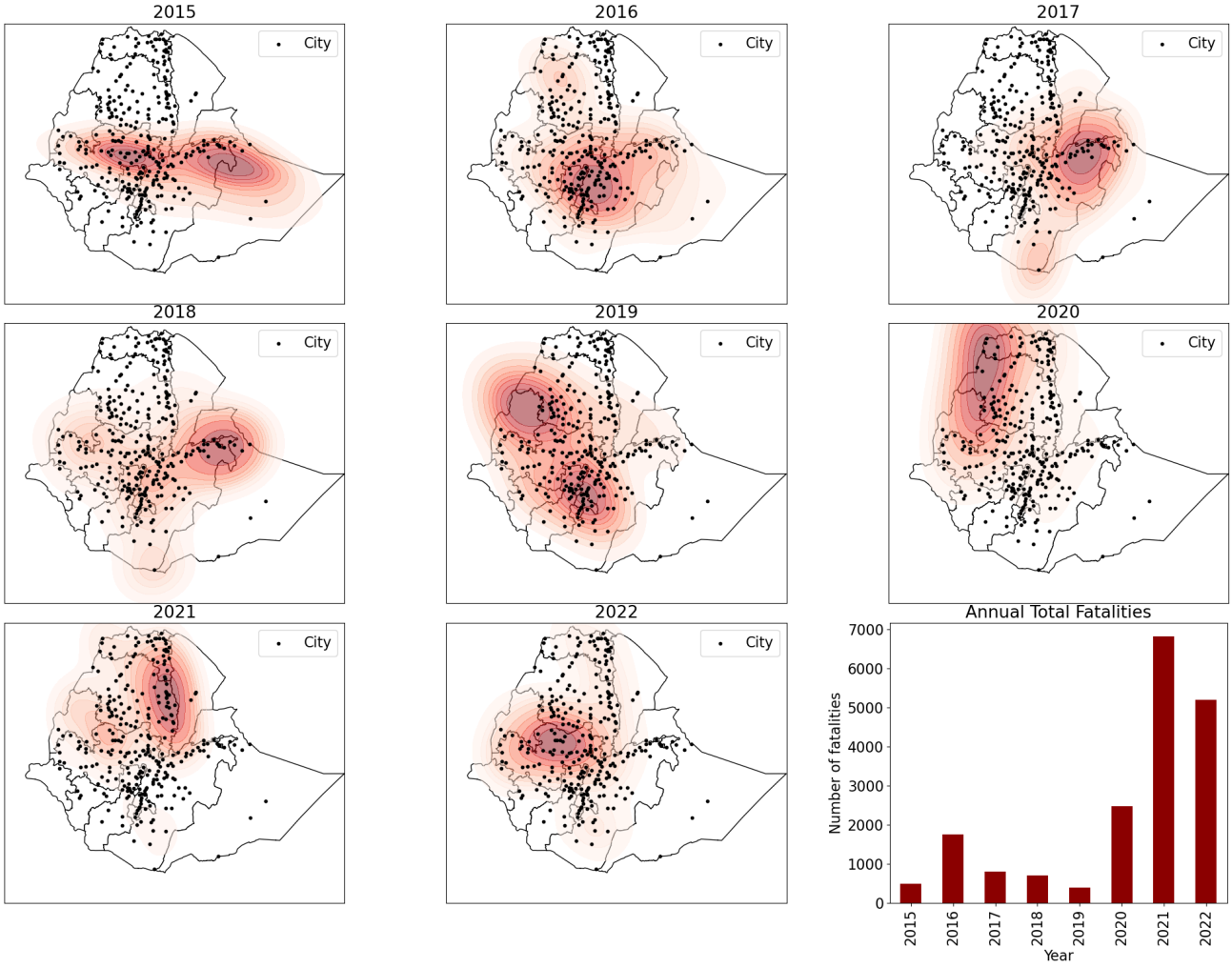
We identify the ethnic groups involved in each ACLED event from the names in the two primary actor fields and all associated-actor fields. The crosswalk combines literal ethnic identifiers in ACLED's actor labels with classifications from the Ethiopia Peace Observatory and the regional concordance used throughout the analysis. Table G1 reports every substring used in the crosswalk and the corresponding assigned group. The group indicators are non-exclusive because the exposure measure records whether an event involves a bank's ethnic group. If an event involves multiple groups, its fatalities enter the exposure measure once for each involved group, and never more than once within the same group.

Table G1: ACLED actor-to-ethnicity crosswalk

Assigned group	Exact case-sensitive actor-name substrings	Classification basis	Fatal events
Panel A: Ethnic-group crosswalk			
Oromo	`Oromo`; `Oromia`; `Omoro`; `Qeerroo`	Literal ethnic and regional identifiers; the EPO actor profile classifies Qeerroo as an Oromo youth movement.	1,263
Tigray	`Tigray`; `Samri`; `Welkait`	Literal ethnic identifiers; the EPO actor profile classifies Samri as a Tigrayan youth group; Welkait follows the project crosswalk.	443
Amhara	`Amhara`; `Fano`	Literal ethnic identifiers; the EPO actor profile classifies Fano as an Amhara militia.	422
Somali	`Somali`; `Garre`; `Gabra`	Literal ethnic identifiers and the project clan crosswalk. Actor labels referring to Somalia rather than Ethiopian Somali groups are excluded.	79
Ogaden	`Ogaden`	Literal clan and organization identifiers, including the Ogaden National Liberation Front; retained as a separate category.	49
Afar	`Afar`	Literal ethnic and regional identifiers.	103
Gumuz	`Gumuz`	Literal ethnic identifiers.	60
SNNP aggregate	`Amaro`; `Konso`; `Guji`; `Sidama`; `Derashe`; `Dirashe`; `Wolayta`; `Wolayita`; `Welayta`; `Gurage`; `Gamo`; `Gofa`; `Silte`; `Burji`; `Kebena`; `Mareko`; `Ale`; `Xonsita`; `Mashole`; `Moseye`; `Kefa`; `Dawero`; `Sheka`; `Konta`; `Dorze`; `Gedeo`; `Koore`; `Meskan`; `Hadiya`; `Shinash`	Groups combined in the project's stable SNNP category using the EPO SNNPR regional profile and the regional concordance used for bank and city ethnicity.	60
Panel B: Event-level coding conventions			
Actor fields	`actor1`; `actor2`; all semicolon-delimited `assoc_actor_1` and `assoc_actor_2` entries	A group is coded as involved when any primary or associated actor contains one of its Panel A strings.	2,705 total
Multiple groups	Retain every group matched within the same event	The indicators are non-exclusive: fatalities from a multi-group event enter each involved group's exposure once, and never more than once within a group.	472
Other actors	No Panel A string match	Federal and foreign forces, generic civilian or protest actors, unidentified actors, and groups outside the crosswalk remain unassigned to a named ethnic group.	716

Notes: The table reports the actor-name crosswalk implemented in `processInterethnicConflict.do`. Counts use the archived ACLED Ethiopia extract for 2015–2022 and include events with positive reported fatalities; because group indicators are non-exclusive, Panel A counts do not sum to the number of events. The main bank-network exposure uses the Oromo, Tigray, Amhara, and SNNP indicators. Sources are the ACLED extract downloaded May 15, 2024; the Ethiopia Peace Observatory actor profiles; and the EPO SNNPR regional profile, accessed March 5, 2025.

Figure G2: Ethnic conflict in Ethiopia



Notes: This figure illustrates the spatial distribution of fatalities due to ethnic conflict in Ethiopia over time. The map on the left presents the distribution of conflict fatalities as a Kernel Density Estimate (KDE) plot, with darker regions indicating areas of higher conflict density. Cities are plotted on the map, represented in black, as well as administrative boundaries of the regions presented in Figure G1. The bottom right bar plot shows the total number of conflict fatalities for each year.

H Attrition and Sample Construction

H.1 Sample Construction

The analysis sample is built from two waves of a branch-level survey of Ethiopian commercial banks, fielded in 2018 and 2022. Table H1 summarises how the final panel is obtained from the original cohort.

2018 baseline. The baseline wave interviewed **1,910** branches spanning 16 banks and 187 cities.

2022 follow-up. In 2022, every branch interviewed in 2018 was targeted for re-interview. Of the 1,910 baseline branches, **996** were successfully re-interviewed. The remaining 914 branches fall into four mutually exclusive categories:

- **138** branches required prior headquarters approval that was not granted in the field window and were therefore not re-interviewed;
- **588** branches could not be contacted (disconnected numbers, wrong numbers, or no response after repeated attempts);
- **151** branches had been confirmed closed between the two waves;
- **37** branches have no recorded 2022 contact status.

These four categories account for the full gap between the 1,910 baseline branches and the 996 successful re-interviews: $996 + 138 + 588 + 151 + 37 = 1{,}910$.

Balanced panel. From the 996 re-interviewed branches, a further 17 are dropped to obtain a balanced panel. One branch is excluded because of a Branch ID issue that we identified as a likely mis-match between the two waves. For the remaining branches we could not determine the exact location, which prevents us from assigning the city-level covariates used in the analysis. The resulting **balanced panel contains 979 branches observed in both waves**, for a total of $979 \times 2 = 1{,}958$ branch-wave observations.

Subsamples. Several specifications split the panel by whether the branch is located in a city whose dominant ethnicity matches that of its bank. This yields **448 branches in bank–city coethnic (BCC) locations** and **531 branches in non–bank–city coethnic (NBCC) locations**, with $448 + 531 = 979$.

H.2 Attrition

We first regress an indicator for whether a branch appears in the 2022 survey on the change in network-level conflict exposure, separately for branches in bank-coethnic and non-bank-coethnic cities. We then estimate a pooled specification with an interaction between conflict exposure and bank–city coethnicity, and finally a simple regression of re-interview status on an indicator for bank–city coethnicity.

Panel A of Table H2 shows that, for branches in non-bank-coethnic cities, changes in conflict exposure have a small and statistically insignificant effect on survey participation. By contrast, in bank-coethnic cities, higher conflict exposure significantly increases attrition. The pooled interaction model in column 3 confirms that the slope is significantly more negative for branches in bank-coethnic cities, while the coefficient on the bank-coethnic-city indicator is positive but statistically insignificant, indicating no baseline difference in attrition between the two types of branches.

Panel B of Table H2 examines the channels of attrition. A one-unit increase in conflict exposure raises the probability that a branch is closed by about 0.5 percentage points in cities that are not coethnic with the bank, with a much larger implied effect, around 8.5 percentage points, in coethnic cities. By contrast, for branches in non-coethnic cities, conflict exposure is associated with a small decline in the probability that a branch requires headquarters approval or cannot be contacted.

Table H1: Sample construction waterfall

Stage	Branches
2018 baseline interviewed	1,910
– 2022 re-interviewed	996
– HQ approval not granted	138
– Could not be contacted	588
– Confirmed closed	151
– No 2022 contact record	37
Re-interviewed in 2022	996
– Dropped: inconsistent identifiers	1
– Dropped: missing city ethnicity	16
Balanced panel (branches)	**979**
Balanced panel (branch-wave obs.)	**1,958**
of which bank–city coethnic (BCC)	448
of which non–bank–city coethnic (NBCC)	531

Table H2: Attrition and Reasons for Non-response

Panel A: Attrition				
	(1) Non-coethnic	(2) Coethnic	(3) Interaction model	(4) Coethnic dummy
Network Exposure	0.013	-0.063***	0.013	
	(0.011)	(0.011)	(0.011)	
Bank and city coethnic			0.026	0.028
			(0.023)	(0.023)
Network Exposure × Bank-city coethnic			-0.075***	
			(0.016)	
Observations	1028	823	1851	1851
R-squared	0.001	0.028	0.014	0.001
Panel B: Reasons for attrition				
	(1) Closed	(2) Requires HQ approval	(3) No contact	
Network Exposure	0.005	-0.010	-0.011	
	(0.004)	(0.007)	(0.010)	
Bank and city coethnic	-0.169***	0.008	0.032	
	(0.025)	(0.014)	(0.028)	
Network Exposure × Bank-city coethnic	0.080***	0.014	-0.012	
	(0.015)	(0.011)	(0.021)	
Observations	1851	1851	1851	

Notes: Panel A reports the coefficient from regressions of the probability of re-interview on the change in log network-level conflict exposure. Columns (1) and (2) report the results for non bank-coethnic and bank-coethnic cities respectively, column (3) interacts conflict with a dummy for whether the bank and city are coethnic. Column (4) regresses re-interview status on an indicator for whether the bank and city are coethnic. Panel B reports average marginal effects from a multinomial logit for the reasons for attrition: the branch is closed, requires HQ approval to participate or cannot be contacted. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. We include all branches for which we determined the location from the 2018 sample. Standard errors are clustered at the branch level.

I Manager Ethnicity Imputation

For a small number of branches in the 2022 survey wave, manager ethnicity could not be directly ascertained through the standard indicators (region of birth, accent, spoken languages, and name). For these cases, we apply the following imputation procedure:

(i). **Carry-forward:** If the same manager was present in both survey waves (identified by matching age, tenure, and gender), we assign the directly observed 2018 ethnicity. These observations are *not* flagged as imputed.

(ii). **Birthplace — same region:** For remaining missing cases where the manager was born in the same region as the branch, we assign the city's dominant ethnicity from the Murdock Ethnographic Atlas.

(iii). **Birthplace — different region:** For remaining missing cases where the manager was born in a different region, we assign the bank's dominant ethnicity.

Steps 2 and 3 are flagged by the variable `ethnicityInferred`. Table I1 reports the number and share of branches affected.

Table I1: Imputation of manager ethnicity

	Count	Share
Total branches in balanced panel	979	
Branches with imputed manager ethnicity	241	24.6%
Assigned city ethnicity (born in same region)	104	10.6%
Assigned bank ethnicity (born elsewhere)	136	13.9%

Notes: Count and share of branches where manager ethnicity in the 2022 wave was imputed using the birthplace rule (steps 2–3). The carry-forward step assigns the directly observed 2018 ethnicity to unchanged managers and is not counted as imputation.

To assess accuracy, we apply the same birthplace rule to observations with known ethnicity. Table I2 reports the results for both survey waves.

Table I3 tests whether the need for imputation is correlated with conflict exposure. We regress the imputation indicator on log conflict intensity, separately for branches in bank-coethnic cities (column 2) and non-bank-coethnic cities (column 3), and test for differential selection by interacting conflict with an indicator for non-bank-coethnic cities (column 4). The coefficients are small and negative, and are statistically insignificant in both subsamples.

Table I2: Accuracy of the ethnicity imputation rule

	N	Correct	Accuracy
Period 1 (2018) — all ethnicities directly observed			
Overall ethnicity match	464	257	55.4%
Born in same region (city rule)	202	106	52.5%
Born elsewhere (bank rule)	262	151	57.6%
Bank coethnic indicator match	464	263	56.7%
City coethnic indicator match	464	270	58.2%
Period 2 (2022) — non-imputed observations only			
Overall ethnicity match	735	424	57.7%
Bank coethnic indicator match	735	454	61.8%
City coethnic indicator match	735	470	63.9%

Notes: Accuracy of the birthplace imputation rule when applied to observations with directly observed manager ethnicity. "Overall ethnicity match" reports the fraction for which the rule assigns the correct ethnic group. "Bank/city coethnic indicator match" reports the fraction for which the imputed value yields the correct coethnic indicator used as dependent variable in the main regressions. Period 1 applies the rule hypothetically to 2018 data where all ethnicities are known. Period 2 applies the rule to 2022 managers whose ethnicity was directly observed (non-imputed).

Table I3: Selection into imputation

	(1) Pooled	(2) BCC	(3) NBCC	(4) Interaction
Network Exposure	-0.030*	-0.044	-0.025	-0.044
	(0.017)	(0.030)	(0.019)	(0.030)
Network Exposure × nonBCE_local				0.019
				(0.035)
R^2	0.003	0.005	0.003	0.017
N	979	448	531	979

Notes: The dependent variable is an indicator equal to one if manager ethnicity was imputed using the birthplace rule (steps 2–3). The sample is restricted to period 2 (2022). Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Column 1 pools all branches; column 2 restricts to branches in bank-coethnic cities; column 3 to branches in non-bank-coethnic cities; column 4 includes the interaction with a non-bank-coethnic city indicator. Robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Finally, Tables I4, I5, and I6 replicate the main manager ethnicity regressions (Table 2), the non-coethnic manager regression (Table J4), and the delegation analysis (Table 3) after excluding all branches with imputed ethnicity. The results are quantitatively and qualitatively similar to the full sample, confirming that the imputation does not drive our findings. The management practices and lending outcomes regressions (not reported) are also robust to this exclusion.

Table I4: Bank-level conflict exposure and manager ethnicity — excluding imputed observations

	Panel A: Branches in Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-0.033	0.006	-0.035	-0.033	0.006	-0.035
	(0.032)	(0.097)	(0.032)	(0.032)	(0.097)	(0.032)
Local Exposure			0.013			0.013
			(0.010)			(0.010)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.659	0.659	0.659	0.659	0.659	0.659
N (Branches)	314	241	314	314	241	314

	Panel B: Branches in Non-Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
Network Exposure	0.047***	0.044**	0.048***	-0.101***	-0.100***	-0.102***
	(0.017)	(0.018)	(0.017)	(0.016)	(0.018)	(0.016)
Local Exposure			-0.014			0.033***
			(0.010)			(0.011)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.196	0.196	0.196	0.755	0.755	0.755
p-value	0.027	0.696	0.021	0.059	0.279	0.060
N (Branches)	424	363	424	424	363	424

Notes: This table replicates Table 2 after excluding branches where manager ethnicity in the 2022 wave was assigned using the birthplace rule. The dependent variable is an indicator equal to one when a branch manager is co-ethnic with the city (columns 1–3) or bank (columns 4–6). Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 5. Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table I5: Bank-level exposure to ethnic conflict and non-coethnic managers — excluding imputed observations

	(1) Bank and city coethnic Non-coethnic manager	(2) Bank and city not coethnic Non-coethnic manager
Network Exposure	0.033	0.054***
	(0.032)	(0.014)
Branch FE	Yes	Yes
Time FE	Yes	Yes
2018 Mean dep. var	0.341	0.050
p-value		0.557
N	314	424

Notes: This table replicates Table J4 after excluding branches where manager ethnicity in the 2022 wave was assigned using the birthplace rule. The dependent variable is an indicator equal to one when the branch manager is coethnic with neither the bank nor the city. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Column 1 restricts the sample to branches in cities coethnic with the bank; column 2 to branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table I6: Bank-level exposure to ethnic conflict and delegation — excluding imputed observations

	Panel A: Ethnic conflict and delegation					
	Bank and city not coethnic			Bank and city coethnic		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-626.2**	-691.2*	-619.3**	100.9	-1077.8	197.9
	(287.0)	(352.3)	(286.1)	(539.5)	(2328.6)	(539.6)
Local Exposure			-74.7			-451.0**
			(138.5)			(183.2)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	1640.9	1672.3	1640.9	1729.4	1794.4	1729.4
p-value	0.234	0.868	0.181	0.234	0.868	0.181
N (Branches)	291	235	291	194	124	194

	Panel B: Manager ethnicity and delegation					
	Panel		Period 1 Only		Period 2 Only	
	(1) Non-coethnic	(2) Coethnic	(3) non-coethnic	(4) Coethnic	(5) non-coethnic	(6) Coethnic
City coethnic manager	-1606.3**	-3156.7***	-144.9	-245.9	-2861.3***	-5171.2***
	(784.4)	(1113.9)	(189.9)	(195.5)	(995.2)	(1793.8)
Branch FE	Yes	Yes	No	No	No	No
Time FE	Yes	Yes	No	No	No	No
Constant	No	No	Yes	Yes	Yes	Yes
2018 Mean dep. var	1640.9	1729.4	1682.8	1636.9	7165.6	7945.6
p-value	0.255	0.255	0.711	0.711	0.260	0.260
N (Branches)	291	194	424	314	291	194

Notes: This table replicates Table 3 after excluding branches where manager ethnicity in the 2022 wave was assigned using the birthplace rule. The dependent variable in both panels is the largest loan a branch manager can approve without headquarters' agreement, measured in thousands of 2018 ETB and winsorized at the 95th percentile within period $\times$ bank-city coethnicity cells. Panel A estimates the effect of Network Exposure on delegation separately for branches where the bank and city are not coethnic (columns 1–3) and where they are coethnic (columns 4–6). Panel B estimates the effect of having a city-coethnic manager on delegation. Odd columns restrict to non-bank-coethnic cities; even columns to bank-coethnic cities. Columns 1–2 use the full panel; columns 3–4 use the 2018 cross-section; columns 5–6 use the 2022 cross-section. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

J Alternative mechanisms tables

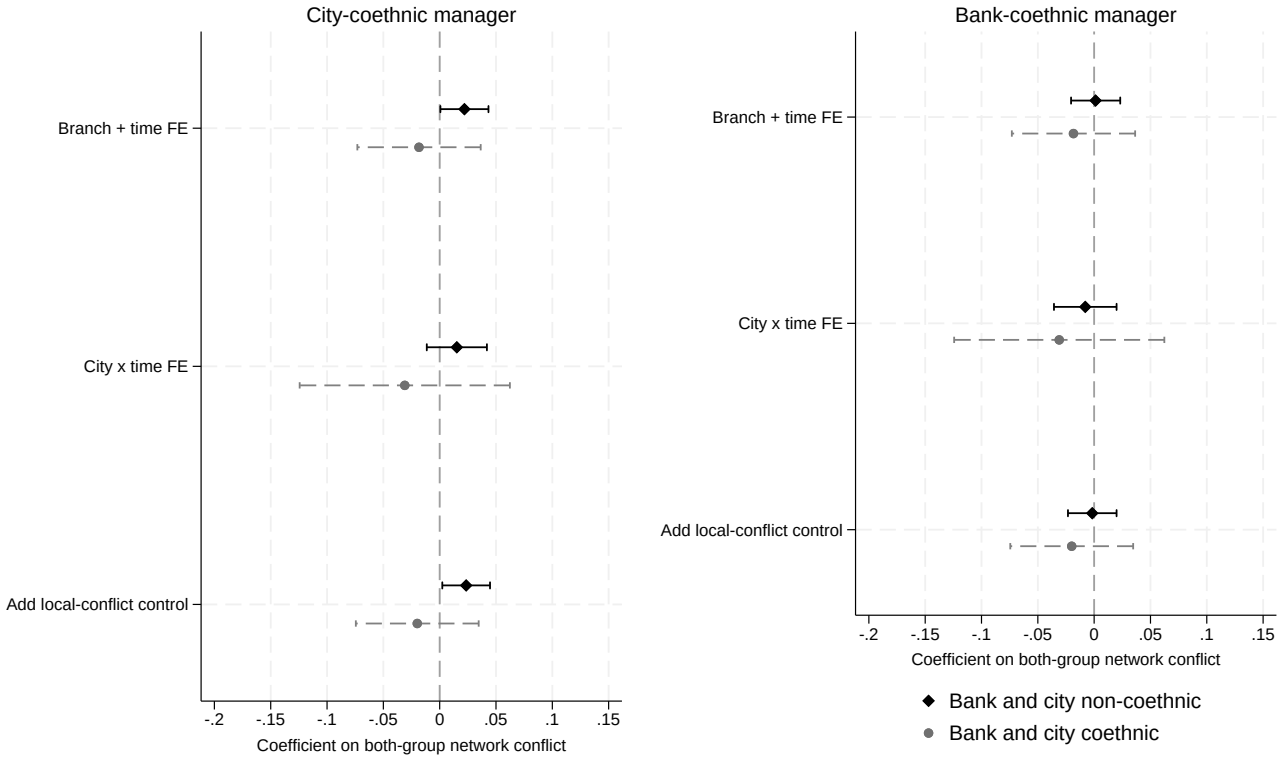
Table J1: Manager ethnicity and delegation by survey period

	Panel		Period 1 Only		Period 2 Only	
	(1)	(2)	(3)	(4)	(5)	(6)
Bank and city coethnic	No	Yes	No	Yes	No	Yes
City coethnic manager	-1526.5**	-3196.2***	-72.7	-320.4*	-2674.6***	-4692.0***
	(751.8)	(1007.0)	(176.7)	(173.0)	(940.9)	(1769.8)
Branch FE	Yes	Yes	No	No	No	No
Time FE	Yes	Yes	No	No	No	No
Constant	No	No	Yes	Yes	Yes	Yes
Mean dep. var.	1684.8	1843.5	1681.7	1681.9	7377.5	8168.2
p-value	0.184	0.184	0.317	0.317	0.314	0.314
N (Branches)	322	230	531	448	322	230

Notes: The dependent variable is the largest loan a branch manager can approve without headquarters' agreement, measured in thousands of 2018 ETB. Values in 2022 are deflated to 2018 prices using annual CPI inflation rates from the World Bank World Development Indicators (indicator FP.CPI.TOTL.ZG), and the dependent variable is winsorized at the 95th percentile within survey-period $\times$ bank-city-coethnicity cells. City coethnic manager is an indicator equal to one when the branch manager shares the city's ethnicity. Columns 1–2 use the full panel and include branch and time fixed effects. Columns 3–4 and 5–6 report separate cross-sectional regressions for 2018 and 2022, respectively. Within each pair, the first column restricts the sample to branches where the bank and city are not coethnic and the second to branches where they are coethnic. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value tests whether the coefficient differs between bank-coethnic and non-bank-coethnic cities within the corresponding sample. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

J.1 Conflict involving both the bank's and the city's ethnic group

Figure J1: Conflict involving both bank and city groups



Notes: Markers report coefficients on network conflict involving both the bank's and the city's ethnic groups; whiskers are 95% confidence intervals based on branch-clustered standard errors. The first specification includes branch and time fixed effects, the second replaces time fixed effects with city-by-time fixed effects, and the third adds local conflict intensity to the first specification.

Table J2: Controlling for local ethnic conflict involving the banks' ethnic group

	Not Bank City Coethnic			Bank city coethnic		
	Bank coethnic	City coethnic	Neither coethnic	Bank coethnic	City coethnic	Neither coethnic
Network Exposure	-0.102***	0.050***	0.051***	-0.009	-0.009	0.009
	(0.016)	(0.017)	(0.013)	(0.027)	(0.027)	(0.027)
Local Coethnic Exposure	0.006	-0.000	-0.005	0.002	0.002	-0.002
	(0.007)	(0.007)	(0.005)	(0.007)	(0.007)	(0.007)
2018 mean dep. var.	0.770	0.185	0.045	0.680	0.680	0.320
N (branches)	531	531	531	453	453	453

This table reports OLS estimates of the effect of bank network conflict intensity on manager ethnicity, controlling for local conflict involving the bank's ethnic group. The dependent variable in each column indicates whether the branch manager is coethnic with the bank, coethnic with the city, or coethnic with neither. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Local coethnic conflict intensity is the log of fatalities involving the bank's ethnicity in the branch's own city. Columns 1–3 restrict to branches where the bank and city dominant ethnicity differ; columns 4–6 restrict to branches where they are the same. All specifications include branch and time fixed effects. Standard errors clustered at the branch level are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table J3: Heterogeneity in the effect of conflict by the nature of local competition

	Panel A: Branches in Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-0.009	-0.005	-0.006	-0.009	-0.005	-0.006
	(0.026)	(0.027)	(0.027)	(0.026)	(0.027)	(0.027)
Many competitors		-0.029			-0.029	
× Network Exposure		(0.045)			(0.045)	
High perceived competition			-0.076			-0.076
× Network Exposure			(0.046)			(0.046)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 mean dep. var.	0.680	0.680	0.680	0.680	0.680	0.680
N (branches)	453	453	453	453	453	453

	Panel B: Branches in Non-Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	0.049***	0.055***	0.050***	-0.096***	-0.101***	-0.096***
	(0.016)	(0.017)	(0.016)	(0.015)	(0.016)	(0.015)
City-coethnic competitor	0.024			0.011		
× Network Exposure	(0.049)			(0.047)		
Many competitors		-0.033			0.035	
× Network Exposure		(0.027)			(0.026)	
High perceived competition			0.029			0.005
× Network Exposure			(0.026)			(0.024)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 mean dep. var.	0.185	0.185	0.185	0.770	0.770	0.770
N (branches)	531	531	531	531	531	531

Notes: The dependent variable is an indicator that equals 1 when a branch's manager is co-ethnic with the ethnic group associated with the city (columns 1–3) or with the ethnic group associated with the branch's bank (columns 4-6). Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 5. The mean (SD) of the within-branch change in log Network Exposure from 2018 to 2022 is 0.953 (1.343). Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the test that the slope on Network Exposure is identical in bank-coethnic and non bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table J4: Bank-level exposure to ethnic conflict and non-coethnic managers

	(1) Bank and city coethnic Non-coethnic manager	(2) Bank and city not coethnic Non-coethnic manager
Network Exposure	0.005	0.047***
	(0.026)	(0.012)
Branch FE	Yes	Yes
Time FE	Yes	Yes
2018 Mean dep. var	0.321	0.045
p-value		0.154
N	448	531

Notes: The dependent variable is an indicator that equals 1 when a branch's manager is co-ethnic with neither the city nor with the ethnic group associated with the branch's bank. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 4. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 5. The mean (SD) of the within-branch change in log Network Exposure from 2018 to 2022 is 0.953 (1.343). Column (1) restricts the sample to branches in cities coethnic with the bank, while column (2) focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the test that the slope on Network Exposure is identical in bank-coethnic and non bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

K Neither Coethnic Results

Table J5: Heterogeneity by the share of local employees in 2018

	Not Bank City Coethnic			Bank city coethnic		
	Bank coethnic	City coethnic	Neither coethnic	Bank coethnic	City coethnic	Neither coethnic
Network Exposure	-0.090***	0.046***	0.044***	-0.008	-0.008	0.008
	(0.016)	(0.016)	(0.013)	(0.025)	(0.025)	(0.025)
High local employees	-0.034	0.022	0.012	0.025	0.025	-0.025
× Network Exposure	(0.025)	(0.027)	(0.022)	(0.044)	(0.044)	(0.044)
2018 mean dep. var.	0.770	0.185	0.045	0.680	0.680	0.320
N (branches)	531	531	531	453	453	453

This table reports OLS estimates of the effect of bank network conflict intensity on manager ethnicity, interacted with an indicator for whether the branch has an above-median share of locally hired employees. The dependent variable in each column indicates whether the branch manager is coethnic with the bank, coethnic with the city, or coethnic with neither. Columns 1–3 restrict to branches where the bank and city dominant ethnicity differ; columns 4–6 restrict to branches where they are the same. All specifications include branch and time fixed effects. Standard errors clustered at the branch level are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table K1: Managers coethnic with neither the bank nor the city

Panel A: Local population share each type of manager would match

	The bank's group	The city's group	The largest third group	Actual manager	N
Bank coethnic	0.178	0.568	0.188	0.178	593
City coethnic	0.162	0.646	0.139	0.646	368
Coethnic with neither	0.157	0.501	0.279	0.263	101
All branches	0.170	0.589	0.180	0.348	1,062

Panel B: Manager composition, ethnic groups defined by the 2007 census

	(1) Bank coethnic (census)	(2) City coethnic (census)	(3) Coethnic w/ neither (census)
Network Exposure	-0.099***	0.061***	0.038***
	(0.015)	(0.016)	(0.012)
Mean dep. var.	0.558	0.368	0.074
N	1,054	1,054	1,054
R^2	0.663	0.637	0.540

Panel C: Manager composition by the size of the largest outside group

	(1) Bank coethnic	(2) City coethnic	(3) Coethnic w/ neither
Network Exposure, no interaction	-0.097***	0.050***	0.047***
	(0.015)	(0.016)	(0.012)
Network Exposure, below cut	-0.095***	0.062***	0.033***
	(0.016)	(0.017)	(0.013)
Network Exposure, above cut	-0.106***	-0.009	0.115***
	(0.031)	(0.035)	(0.031)
Difference (above − below)	-0.011	-0.071*	0.082**
	(0.033)	(0.036)	(0.032)
2018 mean dep. var.	0.770	0.185	0.045
Branch-years above cut	134	134	134
N	1,062	1,062	1,062
Branches	531	531	531

Notes: All panels use bank-city non-coethnic branch-years, where the three manager types are distinct.

Panel A reports how locally rooted each type of manager would be. Rows group branches by the manager they actually have. Columns 1–3 report, for those same branches, the 2007 census population share in their own city of the bank's ethnic group, of the group the Murdock ethnographic atlas assigns to the city, and of the largest group that is neither; column 4 is the share of the manager actually in post. These are census *population* shares, not customers: the survey contains no customer-ethnicity data, the shares come from a single 2007 cross-section, and the SNNP category aggregates more than thirty distinct groups, so an SNNP manager's coethnicity with local residents is an upper bound.

Panel B reports OLS estimates of the effect of network conflict exposure on the ethnicity of the branch manager, with the city's ethnic group taken as the dominant group in the 2007 census rather than the group assigned by the Murdock atlas. Both the sample split and the manager-coethnicity outcomes use that definition; manager and bank ethnicity are unchanged. Redefining the city group flips the bank-city coethnicity classification for 76 branch-periods. The dependent variables are indicators for the manager being coethnic with the bank (column 1), with the city (column 2), or with neither (column 3); the three coefficients sum to zero because the categories partition the sample. Network exposure is the log intensity of branch-weighted exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in equation 4. All columns include branch and wave fixed effects.

Panel C allows the effect of network conflict exposure on manager ethnicity to differ in cities where the largest census group that is coethnic with neither the bank nor the city exceeds 30 percent of the city population (134 of 1,062 branch-years). The dependent variables are indicators for the manager being coethnic with the bank (column 1), with the city (column 2), or with neither (column 3); each coefficient row sums to zero across columns. Each column reports, from one regression, the exposure coefficient without the interaction, the below- and above-cut exposure slopes, and their difference (the interaction coefficient); each estimate carries its own standard error, clustered at the branch level, in parentheses. Manager and city ethnicity are defined as in the main text; all columns include branch and wave fixed effects. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

L AI survey methodology

This appendix describes the AI survey exercise summarized in Section 4.4. We follow John J Horton, Apostolos Filippas and Benjamin S Manning (2023), Lisa P Argyle, Ethan C Busby, Nancy Fulda, Joshua R Gubler, Christopher Rytting and David Wingate (2023), and James Brand, Ayelet Israeli and Donald Ngwe (2023). Figure 4 provides an overview.

L.1 Sample and Vignettes

The sample consists of 200 branches in cities whose dominant ethnic group differs from that associated with the bank, drawn in two independent groups of 100. The first group contains the branches selected for the earlier AI exercise. Each vignette poses a vacancy in 2022 and includes the observed branch location, bank and city ethnic affiliations, the 2018 incumbent's ethnicity and tenure, and actual local conflict in 2019–2022. The model does not see the manager actually appointed in 2022.

L.2 Synthetic CEO and Prompt

For each branch, the model evaluates a city-coethnic candidate and a bank-coethnic candidate together. Network conflict takes two values, 0.6 and 17.5 average fatalities involving the bank's ethnic group near its other branches. These are the 10th and 90th percentiles of the pooled 2018–2022 network-conflict distribution used in the earlier exercise. Within each conflict condition, we present the bank-coethnic candidate first in one conversation and the city-coethnic candidate first in another. The observed local fatality count remains fixed. The sample contains 800 conversations, each with assessments of both candidates, creating 1,600 candidate rows for the assessment regressions.

We construct a system prompt that instructs the LLM to role-play as the bank's CEO. A cached summary of the bank's 2022 annual report supplies context on its branch network, staffing practices, lending, risk management, regional performance, and ethnic references. The prompt then describes how branch managers obtain borrower information, communicate recommendations to headquarters, and exercise approval authority. It gives the same account of coethnic borrower ties for either candidate; a city-coethnic candidate faces those ties on a larger share of local loans. The complete rendered prompts, their hashes, code, and raw responses are retained with the run.

L.3 Survey Questions and Implementation

Each conversation asks a sequence of 20 questions. In the first 17, the synthetic CEO rates both candidates' information acquisition, alignment of recommendations with headquarters, own-group favoritism, safety concerns, the wage premium, expected nonperforming loans, a delegation threshold, and lending outcomes. Favoritism is defined as the percentage of autonomously approved own-group loans that would fail the bank's usual lending standards. Alignment is the percentage of recommendations that would match the bank's recommendation given the same information. The CEO then supplies an appointment probability and a final appointment choice. Three further questions ask for the appointment probability, choice, and delegation threshold if neither candidate favors borrowers of their own group; Appendix Table L1 reports these counterfactual answers.

The conversations use the dated `gpt-5-mini-2025-08-07` model snapshot at temperature 1. The first 100 branches were run in three stages on 8 October 2026; the second 100 were run on 20 October 2026 with the same instrument. Both runs completed all 400 conversations with no conversation-level API failures. Six NPL pairs initially missing from the parsed data were recovered from the first line of their original responses, which clearly states both estimates. One counterfactual final choice remains unparsed.

L.4 Empirical Specification

For a decision Y made in conversation r for branch i under conflict treatment c, we estimate

$$Y_{icr} = \alpha_i + \tau \, \mathrm{High}_c + \omega \, \mathrm{CityFirst}_r + u_{icr}. \tag{a49}$$

The coefficient τ is the within-branch high-minus-low difference, averaged over candidate order. Panel B of Table 7 reports the baseline probability of appointing the city-coethnic candidate, the final choice, and the approval limit assigned to the chosen manager. Table L1 stacks baseline and bias-off answers and adds an indicator for bias off and its interaction with centered high conflict. The bias-off coefficient is therefore its average effect across conflict levels. Appendix Table L2 allows the conflict effect on appointment to differ according to which candidate appears first.

For each candidate assessment, the two candidate rows in a conversation are stacked:

$$Z_{icrm} = \alpha_i + \delta \, \mathrm{City}_m + \theta \, \mathrm{High}_c + \kappa \, \mathrm{City}_m \mathrm{High}_c + e_{icrm}. \tag{a50}$$

The coefficient δ compares city- and bank-coethnic candidates under low conflict; θ is the effect of high conflict for the bank-coethnic candidate, and κ is the additional effect for the city-coethnic candidate. Standard errors in both equations are clustered by branch.

L.5 Results

Table 7 reports the pooled 200-branch estimates. Under high conflict, the synthetic CEO lowers the appointed manager's loan approval limit by 158.4 thousand ETB (s.e. 46.1). Its probability of appointing the city-coethnic candidate increases by 0.022 (s.e. 0.020), and its final choice increases by 0.038 (s.e. 0.029). These appointment effects are imprecise. The authority response agrees in sign with the branch evidence; the appointment response does not reproduce its precision.

For candidate-specific authority, high conflict changes the bank-coethnic candidate's approval limit by -3.4 thousand ETB (s.e. 45.3) and the city-coethnic candidate's by -128.8 thousand ETB, a difference of -125.4 (s.e. 63.6). Candidate ratings show a relative shift toward the city-coethnic candidate in information acquisition ($+4.47$ points) and against that candidate in alignment (-2.96), favoritism ($+3.33$), and expected NPL ($+1.23$ percentage points). High conflict also raises the bank-coethnic candidate's safety concern and wage premium more than the city-coethnic candidate's. These ratings leave both the information–incentive trade-off and differential posting costs as possible influences on the synthetic appointment decision.

Appendix Table L1 reports the answers when own-group favoritism is switched off. The stated appointment probability rises by 0.409 (s.e. 0.015), and the synthetic CEO chooses the city-coethnic candidate in 99.7 percent of final decisions, essentially every case. The appointed manager's approval limit rises by 1,181.8

Table L1: Appointment and Branch Delegation When Own-Group Favoritism Is Switched Off

	(1) P(appoint city-coethnic)	(2) Appoints city-coethnic	(3) Delegation, appointed manager
High conflict	0.022	0.037	-158.375***
	(0.020)	(0.029)	(46.122)
Bias off	0.409***	0.546***	1181.785***
	(0.015)	(0.023)	(27.362)
High conflict × bias off	-0.020	-0.037	100.069**
	(0.019)	(0.029)	(49.691)
Branch FE	Yes	Yes	Yes
Dep. var. mean	0.683	0.724	1698.093
N	1,600	1,599	1,599
R^2	0.563	0.512	0.566

Notes: The table stacks the baseline and bias-off answers from each of 800 conversations at 200 branches. The counterfactual asks the synthetic CEO to suppose that neither candidate favors coethnic borrowers and both make every lending decision purely in the bank's interest. High conflict is centered, so "Bias off" is its average effect across the two conflict levels. All regressions include branch fixed effects; standard errors are clustered by branch. Appointment probabilities are on a 0–1 scale and delegation is in thousands of ETB. One counterfactual final choice did not parse. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

thousand ETB (s.e. 27.4). Appendix Table L2 shows that the conflict effect on appointment does not differ significantly according to candidate order: the interaction estimates are -0.047 (s.e. 0.039) for appointment probability and -0.045 (s.e. 0.057) for final choice.

Table L2: Appointment Effects by Candidate Order

	(1) P(appoint city-coethnic)	(2) Appoints city-coethnic
High conflict, bank-coethnic first	0.045*	0.060
	(0.027)	(0.040)
City-coethnic first	0.045	0.125***
	(0.030)	(0.043)
High conflict × city-coethnic first	-0.047	-0.045
	(0.039)	(0.057)
Branch FE	Yes	Yes
N	800	800

Notes: Each regression uses one observation per conversation and includes branch fixed effects. "City-coethnic first" indicates that the city-coethnic candidate is presented before the bank-coethnic candidate; the bank-coethnic-first order is the reference category. The interaction tests whether the effect of high conflict differs by order. Its p-value is 0.233 for appointment probability and 0.430 for final choice. Standard errors clustered by branch are in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

References

Aghion, Philippe, and Jean Tirole. 1997. "Formal and Real Authority in Organizations." *Journal of Political Economy*, 105(1): 1–29.

Alonso, Ricardo, Wouter Dessein, and Niko Matouschek. 2008. "When Does Coordination Require Centralization?" *American Economic Review*, 98(1): 145–79.

Argyle, Lisa P, Ethan C Busby, Nancy Fulda, Joshua R Gubler, Christopher Rytting, and David Wingate. 2023. "Out of one, many: Using language models to simulate human samples." *Political Analysis*, 31(3): 337–351.

Bertrand, Marianne, and Sendhil Mullainathan. 2001. "Do people mean what they say? Implications for subjective survey data." *American Economic Review*, 91(2): 67–72.

Brand, James, Ayelet Israeli, and Donald Ngwe. 2023. "Using LLMs for market research." *Harvard business school marketing unit working paper*, , (23-062).

Cameron, A Colin, Jonah B Gelbach, and Douglas L Miller. 2008. "Bootstrap-based improvements for inference with clustered errors." *The review of economics and statistics*, 90(3): 414–427.

Crawford, Vincent P., and Joel Sobel. 1982. "Strategic Information Transmission." *Econometrica*, 50(6): 1431–1451.

Dessein, Wouter. 2002. "Authority and Communication in Organizations." *The Review of Economic Studies*, 69(4): 811–838.

Fisman, Raymond, Arkodipta Sarkar, Janis Skrastins, and Vikrant Vig. 2020. "Experience of communal conflicts and intergroup lending." *Journal of Political Economy*, 128(9): 3346–3375.

Fisman, Raymond, Daniel Paravisini, and Vikrant Vig. 2017. "Cultural proximity and loan outcomes." *American Economic Review*, 107(2): 457–492.

Gennaioli, Nicola, Nicola Limodio, and Francesco Strobbe. 2019. "Branch management and firm credit."

Horton, John J, Apostolos Filippas, and Benjamin S Manning. 2023. "Large language models as simulated economic agents: What can we learn from homo silicus?" National Bureau of Economic Research.

Liberti, Jose M., and Atif R. Mian. 2009. "Estimating the Effect of Hierarchies on Information Use." *The Review of Financial Studies*, 22(10): 4057–4090.

Murdock, George Peter. 1959. "Africa its peoples and their culture history."

Nunn, Nathan. 2008. "The Long-Term Effects of Africa's Slave Trades." *Quarterly Journal of Economics*, 123(1): 139–176.

Roodman, David, Morten Ørregaard Nielsen, James G MacKinnon, and Matthew D Webb. 2019. "Fast and wild: Bootstrap inference in Stata using boottest." *The Stata Journal*, 19(1): 4–60.

Stein, Jeremy C. 2002. "Information Production and Capital Allocation: Decentralized versus Hierarchical Firms." *The Journal of Finance*, 57(5): 1891–1921.

Strassen, Volker. 1965. "The existence of probability measures with given marginals." *The Annals of Mathematical Statistics*, 36(2): 423–439.